

In [137...

```
import os
import time
from pathlib import Path

import numpy as np
import pandas as pd
import requests

os.environ["CENSUS_API_KEY"] = "b1278b466e026ad679a62a03dd4b044bb8b36e7c"
CENSUS_API_KEY = os.getenv("CENSUS_API_KEY")
print("Key loaded:", CENSUS_API_KEY is not None)

pd.set_option("display.max_columns", 200)
pd.set_option("display.width", 200)
pd.set_option("display.max_colwidth", 120)

# -----
# Paths / config
# -----
DATA_DIR = Path("data")
RAW_DIR = DATA_DIR / "raw"
OUT_DIR = DATA_DIR / "processed"
RAW_DIR.mkdir(parents=True, exist_ok=True)
OUT_DIR.mkdir(parents=True, exist_ok=True)

# Update this path if needed
IGS_PATH = r"C:\Users\jabba\Desktop\Code\machine_learning\AUC_mastercard_cha

# Tract selection for focused EDA / testing
TARGET_TRACTS = None
# Example:
# TARGET_TRACTS = ["13089023301", "13089023405", "13089023302"]

# If TARGET_TRACTS is set, the code will infer state FIPS from those tracts
SAVE_INTERMEDIATE = True

# ACS 5-year currently available through 2024
ACS_MAX_AVAILABLE_YEAR = 2024
FILL_2025_WITH_2024_ACS = False # optional, keep False by default
```

Key loaded: True

Cell 2 — IGS load / clean helpers

In [138...

```
IGS_STRING_COLS = {
    "County",
    "State",
    "BENCHMARK",
    "Census Tract FIPS code",
    "geoid",
}

IGS_ALIAS_MAP = {
    "Inclusive Growth Score": "igs_total",
```

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    "Growth": "igs_growth",
    "Inclusion": "igs_inclusion",
    "Place": "igs_place",
    "Economy": "igs_economy",
    "Community": "igs_community",
    "Place Growth": "igs_place_growth",
    "Place Inclusion": "igs_place_inclusion",
    "Economy Growth": "igs_economy_growth",
    "Economy Inclusion": "igs_economy_inclusion",
    "Community Growth": "igs_community_growth",
    "Community Inclusion": "igs_community_inclusion",
}

def dedupe_columns(cols):
    seen = {}
    out = []
    for c in cols:
        c = str(c).strip()
        if c not in seen:
            seen[c] = 0
            out.append(c)
        else:
            seen[c] += 1
            out.append(f"{c}.{seen[c]}")
    return out

def _find_header_row(df_raw, required_tokens=("Census Tract FIPS", "Year")):
    tokens = [t.lower() for t in required_tokens]
    for i in range(min(len(df_raw), 50)):
        row = df_raw.iloc[i].astype(str).str.lower().tolist()
        if all(any(tok in cell for cell in row) for tok in tokens):
            return i
    raise ValueError("Could not find the real IGS header row.")

def normalize_geoid_series(series: pd.Series) -> pd.Series:
    s = series.astype("string").str.strip()

    numeric = pd.to_numeric(s, errors="coerce")
    numeric_mask = numeric.notna()

    s = s.copy()
    s.loc[numeric_mask] = numeric.loc[numeric_mask].astype("Int64").astype("string")

    s = s.str.replace(r"\.0$", "", regex=True)
    s = s.str.replace(r"\D", "", regex=True)
    s = s.str.zfill(11)
    s = s.where(s.str.fullmatch(r"\d{11}"), pd.NA)

    return s

def filter_tracts(df: pd.DataFrame, tracts=None, geoid_col="geoid") -> pd.DataFrame:
    if not tracts:

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        return df.copy()

    tract_set = {str(t).zfill(11) for t in tracts}
    out = df[df[geoid_col].astype("string").isin(tract_set)].copy()
    return out

def safe_save_parquet(df: pd.DataFrame, path: Path):
    out = df.copy()

    for c in out.columns:
        if out[c].dtype == "object":
            out[c] = out[c].astype("string")

    out.to_parquet(path, index=False, engine="pyarrow")

def load_igs_any(path: str) -> pd.DataFrame:
    p = Path(path)

    if p.suffix.lower() in [".xlsx", ".xls"]:
        raw = pd.read_excel(p, header=None)
    else:
        raw = pd.read_csv(p, header=None, low_memory=False)

    hdr_i = _find_header_row(raw, required_tokens=("Census Tract FIPS", "Year"))
    header = dedupe_columns(raw.iloc[hdr_i].tolist())

    df = raw.iloc[hdr_i + 1 :].copy()
    df.columns = header
    df = df.dropna(how="all").reset_index(drop=True)

    # Drop any duplicated header rows that still appear in the body
    if "Census Tract FIPS code" in df.columns:
        df = df[
            df["Census Tract FIPS code"].astype(str).str.strip().str.lower()
            .copy()

    # Normalize GEOID + year
    df["geoid"] = normalize_geoid_series(df["Census Tract FIPS code"])
    df["year"] = pd.to_numeric(df["Year"], errors="coerce")

    # Convert non-ID columns to numeric where possible
    for c in df.columns:
        if c not in IGS_STRING_COLS:
            df[c] = pd.to_numeric(df[c], errors="coerce")

    # Friendly aliases for easier modeling later
    for old_col, new_col in IGS_ALIAS_MAP.items():
        if old_col in df.columns:
            df[new_col] = pd.to_numeric(df[old_col], errors="coerce")

    if "Is an Opportunity Zone" in df.columns:
        df["is_opp_zone"] = pd.to_numeric(df["Is an Opportunity Zone"], errors="coerce")

    if "URBAN CODE" in df.columns:

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df["urban_code"] = pd.to_numeric(df["URBAN CODE"], errors="coerce")

if "BENCHMARK" in df.columns:
    df["benchmark"] = df["BENCHMARK"].astype("string")

# Final cleanup
df = df.dropna(subset=["geoid", "year"]).copy()
df["year"] = df["year"].astype(int)
df = df.sort_values(["geoid", "year"]).reset_index(drop=True)

return df

```

Cell 3 — Load and save clean IGS

```

In [139... igs = load_igs_any(IGS_PATH)
igs = filter_tracts(igs, TARGET_TRACTS)

if SAVE_INTERMEDIATE:
    safe_save_parquet(igs, OUT_DIR / "igs_clean.parquet")

print("IGS shape:", igs.shape)
print("IGS years:", sorted(igs["year"].dropna().unique().tolist())[:10], "...")
print()
print(igs[["geoid", "year", "igs_total"]].head())
print()
print(igs[["igs_total", "igs_place", "igs_economy", "igs_community"]].describe())
print()
print("Top missingness:")
print(igs.isna().mean().sort_values(ascending=False).head(20))

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IGS shape: (765288, 87)
IGS years: [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025] ... [2023, 2024, 2025]
```

```

      geoid  year  igs_total
0  01001020100  2017      47.0
1  01001020100  2018      52.0
2  01001020100  2019      46.0
3  01001020100  2020      46.0
4  01001020100  2021      38.0

```

	igs_total	igs_place	igs_economy	igs_community
count	757582.000000	761912.000000	764294.000000	762407.000000
mean	50.052242	49.848921	50.328149	50.148915
std	10.252555	12.453479	13.532944	13.744805
min	0.000000	0.000000	0.000000	0.000000
25%	44.000000	42.000000	42.000000	40.000000
50%	50.000000	50.000000	51.500000	50.000000
75%	57.900000	58.000000	60.000000	60.000000
max	84.000000	100.000000	94.000000	100.000000

```

Top missingness:
Is an Opportunity Zone                1.000000
Spend Growth Base, %                  1.000000
is_opp_zone                           1.000000
Spending per Capita Base, %           1.000000
Spending per Capita Tract, %           1.000000
Spend Growth Tract, %                  1.000000
Early Education Enrollment Tract, %    0.176266
Early Education Enrollment Score        0.176266
Minority/Women Owned Businesses Base, % 0.159723
Minority/Women Owned Businesses Score   0.159723
Minority/Women Owned Businesses Tract, % 0.159723
Spend Growth Score                     0.099624
Spending per Capita Score               0.099624
Residential Real Estate Value Score     0.050105
Residential Real Estate Value Tract, %  0.050105
Travel Time to Work Tract, %            0.024643
Travel Time to Work Score               0.024643
Affordable Housing Score                 0.019497
Affordable Housing Tract, %              0.019497
Female Above Poverty Tract, %            0.018882
dtype: float64

```

Cell 4 — ACS variables to pull

```

In [140... ACS_VAR_GROUPS = {
    "population_age": [
        "B01001_001E", # total pop
        "B01001_003E", "B01001_004E", "B01001_005E", "B01001_006E", # male
        "B01001_027E", "B01001_028E", "B01001_029E", "B01001_030E", # female
        "B01001_020E", "B01001_021E", "B01001_022E", "B01001_023E", "B01001_
        "B01001_044E", "B01001_045E", "B01001_046E", "B01001_047E", "B01001_
    ],
    "race": [

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        "B02001_001E",
        "B02001_002E",
        "B02001_003E",
        "B02001_005E",
        "B02001_008E",
    ],
    "education": [
        "B15003_001E",
        "B15003_022E", "B15003_023E", "B15003_024E", "B15003_025E",
    ],
    "employment": [
        "B23025_001E",
        "B23025_002E",
        "B23025_003E",
        "B23025_005E",
    ],
    "income_poverty": [
        "B19013_001E", # median household income
        "B19301_001E", # per-capita income
        "B19083_001E", # gini
        "B17001_001E", # poverty universe
        "B17001_002E", # below poverty
    ],
    "commute": [
        "B08303_001E",
        "B08303_002E", "B08303_003E", "B08303_004E", "B08303_005E",
        "B08303_006E", "B08303_007E", "B08303_008E",
    ],
    "internet": [
        "B28002_001E",
        "B28002_002E",
    ],
    "housing": [
        "B25002_001E", "B25002_002E", "B25002_003E", # total / occupied /
        "B25003_001E", "B25003_002E", "B25003_003E", # tenure
        "B25064_001E", # median gross rent
        "B25077_001E", # median home value
    ],
    "rent_burden": [
        "B25070_001E",
        "B25070_002E", "B25070_003E", "B25070_004E", "B25070_005E", "B25070_
    ],
    "owner_cost_burden": [
        "B25091_001E",
        "B25091_003E", "B25091_004E", "B25091_005E", "B25091_006E", "B25091_
        "B25091_014E", "B25091_015E", "B25091_016E", "B25091_017E", "B25091_
    ],
    "early_education": [
        "B14003_004E", "B14003_013E", "B14003_032E", "B14003_041E",
    ],
    "health_insurance": [
        "B27001_001E",
        "B27001_005E", "B27001_008E", "B27001_011E", "B27001_014E", "B27001_
        "B27001_020E", "B27001_023E", "B27001_026E", "B27001_029E",
        "B27001_033E", "B27001_036E", "B27001_039E", "B27001_042E", "B27001_
        "B27001_048E", "B27001_051E", "B27001_054E", "B27001_057E",
    ],

```

```

    ],
}

ACS_VARS = sorted({v for group in ACS_VAR_GROUPS.values() for v in group})
print("ACS variable count:", len(ACS_VARS))

```

ACS variable count: 98

Cell 5 — ACS API helpers

```

In [141]: def chunks(lst, n):
    for i in range(0, len(lst), n):
        yield lst[i : i + n]

def infer_state_fips_from_tracts(tracts):
    if not tracts:
        return None
    return sorted({str(t).zfill(11)[:2] for t in tracts})

def resolve_acs_years(igs_years, max_available_year=ACS_MAX_AVAILABLE_YEAR):
    igs_years = sorted({int(y) for y in igs_years if pd.notna(y)})
    acs_years = [y for y in igs_years if y <= max_available_year]
    return acs_years

def get_state_fips(year: int, api_key: str) -> list[str]:
    url = f"https://api.census.gov/data/{year}/acs/acs5"
    params = {"get": "NAME", "for": "state:*", "key": api_key}

    r = requests.get(url, params=params, timeout=60)
    r.raise_for_status()

    data = r.json()
    header, rows = data[0], data[1:]
    df = pd.DataFrame(rows, columns=header)

    return sorted(df["state"].unique().tolist())

def census_get_tracts(year: int, state_fips: str, vars_: list[str], api_key: str):
    base = f"https://api.census.gov/data/{year}/acs/acs5"
    get_str = "NAME," + ",".join(vars_)

    params = [
        ("get", get_str),
        ("for", "tract:*"),
        ("in", f"state:{state_fips}"),
        ("in", "county:*"),
        ("key", api_key),
    ]

    r = requests.get(base, params=params, timeout=120)
    r.raise_for_status()

```

```

data = r.json()
header, rows = data[0], data[1:]
df = pd.DataFrame(rows, columns=header)
df["year"] = year

return df

def to_numeric_safe(df: pd.DataFrame, exclude=("NAME", "state", "county", "t
out = df.copy()
for c in out.columns:
    if c in exclude:
        continue
    out[c] = pd.to_numeric(out[c], errors="coerce")
return out

ACS_SENTINELS = {-666666666, -333333333, -222222222}

def clean_acs_values(df: pd.DataFrame, exclude=("NAME", "state", "county", "
out = df.copy()
for c in out.columns:
    if c in exclude:
        continue
    out[c] = pd.to_numeric(out[c], errors="coerce")
    out[c] = out[c].replace(list(ACS_SENTINELS), np.nan)
return out

def download_acs_all(
    years: list[int],
    api_key: str,
    state_fips_filter: list[str] | None = None,
    max_vars_per_call: int = 45,
    sleep_seconds: float = 0.25,
) -> pd.DataFrame:
    if not api_key:
        raise ValueError("Missing CENSUS_API_KEY. Put it in your environment

    all_parts = []

    for year in years:
        year_out = RAW_DIR / f"acs5_tract_{year}.parquet"

        if year_out.exists():
            print(f"[cache] {year_out.name}")
            df_year = pd.read_parquet(year_out)
            all_parts.append(df_year)
            continue

        states = state_fips_filter if state_fips_filter else get_state_fips(
            print(f"[download] year={year} | states={len(states)}")

        state_frames = []

        for st in states:

```



```

merged_state = None

for var_chunk in chunks(ACS_VARS, max_vars_per_call):
    df_chunk = census_get_tracts(year, st, var_chunk, api_key)
    df_chunk = clean_acs_values(df_chunk)

    keys = ["state", "county", "tract", "NAME", "year"]
    if merged_state is None:
        merged_state = df_chunk
    else:
        merged_state = merged_state.merge(df_chunk, on=keys, how='outer')

    time.sleep(sleep_seconds)

state_frames.append(merged_state)

df_year = pd.concat(state_frames, ignore_index=True)

if SAVE_INTERMEDIATE:
    safe_save_parquet(df_year, year_out)

all_parts.append(df_year)

acs = pd.concat(all_parts, ignore_index=True)
return acs

```

Cell 6 — ACS feature engineering

In [142...

```

def safe_div(n, d):
    n = np.asarray(n, dtype="float64")
    d = np.asarray(d, dtype="float64")

    out = np.full_like(n, np.nan, dtype="float64")
    valid = (~np.isnan(d)) & (d != 0)
    np.divide(n, d, out=out, where=valid)
    return out

def make_features(acs: pd.DataFrame) -> pd.DataFrame:
    acs = acs.copy()

    acs["state"] = acs["state"].astype(str).str.zfill(2)
    acs["county"] = acs["county"].astype(str).str.zfill(3)
    acs["tract"] = acs["tract"].astype(str).str.zfill(6)
    acs["geoid"] = acs["state"] + acs["county"] + acs["tract"]

    out = pd.DataFrame({
        "geoid": acs["geoid"],
        "year": acs["year"],
    })

    # -----
    # Population / age
    # -----
    out["pop_total"] = acs["B01001_001E"]

```

```

out["pop_under18"] = (
    acs["B01001_003E"] + acs["B01001_004E"] + acs["B01001_005E"] + acs["
    acs["B01001_027E"] + acs["B01001_028E"] + acs["B01001_029E"] + acs["
)
out["share_under18"] = safe_div(out["pop_under18"], out["pop_total"])

out["pop_65plus"] = (
    acs["B01001_020E"] + acs["B01001_021E"] + acs["B01001_022E"] + acs["
    acs["B01001_044E"] + acs["B01001_045E"] + acs["B01001_046E"] + acs["
)
out["share_65plus"] = safe_div(out["pop_65plus"], out["pop_total"])

# -----
# Race shares
# -----
out["share_white"] = safe_div(acs["B02001_002E"], acs["B02001_001E"])
out["share_black"] = safe_div(acs["B02001_003E"], acs["B02001_001E"])
out["share_asian"] = safe_div(acs["B02001_005E"], acs["B02001_001E"])
out["share_two_plus"] = safe_div(acs["B02001_008E"], acs["B02001_001E"])

# -----
# Education
# -----
ba_plus = acs["B15003_022E"] + acs["B15003_023E"] + acs["B15003_024E"] +
out["ba_plus_share_25p"] = safe_div(ba_plus, acs["B15003_001E"])

# -----
# Employment / economy
# -----
out["lfpr_16p"] = safe_div(acs["B23025_002E"], acs["B23025_001E"])
out["unemp_rate"] = safe_div(acs["B23025_005E"], acs["B23025_003E"])

out["median_household_income"] = acs["B19013_001E"]
out["per_capita_income"] = acs["B19301_001E"]
out["gini"] = acs["B19083_001E"]
out["poverty_rate"] = safe_div(acs["B17001_002E"], acs["B17001_001E"])

# -----
# Commute / internet
# -----
commute_under35 = (
    acs["B08303_002E"] + acs["B08303_003E"] + acs["B08303_004E"] + acs["
    acs["B08303_006E"] + acs["B08303_007E"] + acs["B08303_008E"]
)
out["commute_under35_share"] = safe_div(commute_under35, acs["B08303_001
out["internet_sub_share"] = safe_div(acs["B28002_002E"], acs["B28002_001

# -----
# Housing / affordability
# -----
out["housing_units_total"] = acs["B25002_001E"]
out["occupied_units"] = acs["B25002_002E"]
out["vacant_units"] = acs["B25002_003E"]
out["occupied_share"] = safe_div(acs["B25002_002E"], acs["B25002_001E"])
out["vacancy_rate"] = safe_div(acs["B25002_003E"], acs["B25002_001E"])

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out["owner_share"] = safe_div(acs["B25003_002E"], acs["B25003_001E"])
out["renter_share"] = safe_div(acs["B25003_003E"], acs["B25003_001E"])

out["median_gross_rent"] = acs["B25064_001E"]
out["median_home_value"] = acs["B25077_001E"]

rent_affordable = (
    acs["B25070_002E"] + acs["B25070_003E"] + acs["B25070_004E"] +
    acs["B25070_005E"] + acs["B25070_006E"]
)

owner_affordable = (
    acs["B25091_003E"] + acs["B25091_004E"] + acs["B25091_005E"] + acs["
    acs["B25091_014E"] + acs["B25091_015E"] + acs["B25091_016E"] + acs["
)

denom_housing_cost = acs["B25070_001E"] + acs["B25091_001E"]
out["affordable_housing_share"] = safe_div(rent_affordable + owner_affor

# -----
# Community / health
# -----
# Early education proxy: enrolled 3-4 year olds over under-5 population
enrolled_3_4 = acs["B14003_004E"] + acs["B14003_013E"] + acs["B14003_032
pop_under5 = acs["B01001_003E"] + acs["B01001_027E"]
out["early_ed_enroll_share"] = safe_div(enrolled_3_4, pop_under5)

uninsured = (
    acs["B27001_005E"] + acs["B27001_008E"] + acs["B27001_011E"] + acs["
    acs["B27001_020E"] + acs["B27001_023E"] + acs["B27001_026E"] + acs["
    acs["B27001_033E"] + acs["B27001_036E"] + acs["B27001_039E"] + acs["
    acs["B27001_048E"] + acs["B27001_051E"] + acs["B27001_054E"] + acs["
)
out["insured_share"] = 1.0 - safe_div(uninsured, acs["B27001_001E"])

# -----
# Growth features
# -----
out = out.sort_values(["geoid", "year"]).reset_index(drop=True)

growth_cols = [
    "median_household_income",
    "per_capita_income",
    "occupied_units",
    "median_home_value",
    "median_gross_rent",
]

for col in growth_cols:
    out[f"{col}_growth"] = out.groupby("geoid")[col].pct_change(fill_met

return out

```

Cell 7 — Merge helpers

```

In [143... def prepare_acs_raw_for_merge(acs: pd.DataFrame) -> pd.DataFrame:
    acs_raw = acs.copy()

    acs_raw["state"] = acs_raw["state"].astype(str).str.zfill(2)
    acs_raw["county"] = acs_raw["county"].astype(str).str.zfill(3)
    acs_raw["tract"] = acs_raw["tract"].astype(str).str.zfill(6)
    acs_raw["geoid"] = acs_raw["state"] + acs_raw["county"] + acs_raw["tract"]

    # one row per tract-year expected
    acs_raw = acs_raw.drop_duplicates(subset=["geoid", "year"]).copy()

    return acs_raw

def merge_igs_acs_raw(igs: pd.DataFrame, acs_raw: pd.DataFrame) -> pd.DataFrame:
    merged = igs.merge(acs_raw, on=["geoid", "year"], how="left", validate="m:m")
    return merged

def merge_igs_acs_full(
    igs: pd.DataFrame,
    acs_raw: pd.DataFrame,
    feats: pd.DataFrame,
) -> pd.DataFrame:
    merged = (
        igs
        .merge(acs_raw, on=["geoid", "year"], how="left", validate="m:m")
        .merge(feats, on=["geoid", "year"], how="left", validate="m:m")
    )
    return merged

```

```

In [144... # for year in range(2017, 2025):
#     p = RAW_DIR / f"acs5_tract_{year}.parquet"
#     if p.exists():
#         p.unlink()
#         print("deleted", p.name)

```

Cell 8 — Run ACS pull, feature engineering, and merge

```

In [145... igs_years = sorted(igs["year"].dropna().unique().tolist())
acs_years = resolve_acs_years(igs_years, max_available_year=ACS_MAX_AVAILABLE_YEAR)
state_fips_filter = infer_state_fips_from_tracts(TARGET_TRACTS)

print("IGS years in file:", igs_years)
print("ACS years to pull:", acs_years)
print("State filter from target tracts:", state_fips_filter)

acs = download_acs_all(
    years=acs_years,
    api_key=CENSUS_API_KEY,
    state_fips_filter=state_fips_filter,
    max_vars_per_call=45,
    sleep_seconds=0.25,
)

```

```

)

acs_raw = prepare_acs_raw_for_merge(acs)
feats = make_features(acs)
feats = filter_tracts(feats, TARGET_TRACTS)

# keep only IGS years that exist in ACS-derived tables
valid_years = sorted(feats["year"].dropna().unique())
model_igs = igs[igs["year"].isin(valid_years)].copy()

igs_x_acs_raw = merge_igs_acs_raw(model_igs, acs_raw)
model_df_full = merge_igs_acs_full(model_igs, acs_raw, feats)

if SAVE_INTERMEDIATE:
    safe_save_parquet(acs, OUT_DIR / "acs_raw.parquet")
    safe_save_parquet(acs_raw, OUT_DIR / "acs_raw_with_geoid.parquet")
    safe_save_parquet(feats, OUT_DIR / "acs_features.parquet")
    safe_save_parquet(igs_x_acs_raw, OUT_DIR / "igs_x_acs_raw.parquet")
    safe_save_parquet(model_df_full, OUT_DIR / "igs_x_acs_full.parquet")

print("ACS raw shape:", acs.shape)
print("ACS raw+geoid shape:", acs_raw.shape)
print("ACS features shape:", feats.shape)
print("IGS x ACS raw shape:", igs_x_acs_raw.shape)
print("Full merged model_df_full shape:", model_df_full.shape)

```

IGS years in file: [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025]

ACS years to pull: [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]

State filter from target tracts: None

[cache] acs5_tract_2017.parquet

[cache] acs5_tract_2018.parquet

[cache] acs5_tract_2019.parquet

[cache] acs5_tract_2020.parquet

[cache] acs5_tract_2021.parquet

[cache] acs5_tract_2022.parquet

[cache] acs5_tract_2023.parquet

[cache] acs5_tract_2024.parquet

ACS raw shape: (648952, 103)

ACS raw+geoid shape: (648952, 104)

ACS features shape: (648952, 37)

IGS x ACS raw shape: (680256, 189)

Full merged model_df_full shape: (680256, 224)

validation cell right after Cell 8

```

In [146... # choose the final EDA panel
eda_df = model_df_full.copy()

print("EDA shape:", eda_df.shape)
print("EDA years:", sorted(eda_df["year"].dropna().unique().tolist()))
print("EDA unique tracts:", eda_df["geoid"].nunique())

# required ACS raw columns
required_acs_raw = ["NAME", "state", "county", "tract"] + ACS_VARS
missing_acs_raw = [c for c in required_acs_raw if c not in eda_df.columns]

```

```

# required engineered columns
required_feats = [c for c in feats.columns if c not in ["geoid", "year"]]
missing_feats = [c for c in required_feats if c not in eda_df.columns]

# all original IGS columns
missing_igs = [c for c in igs.columns if c not in eda_df.columns]

print("Missing raw ACS columns:", missing_acs_raw[:20], "count =", len(missing_acs_raw))
print("Missing engineered feature columns:", missing_feats[:20], "count =", len(missing_feats))
print("Missing IGS columns:", missing_igs[:20], "count =", len(missing_igs))

print("Duplicate geoid-year rows:", eda_df.duplicated(subset=["geoid", "year"]).sum())

# quick sample
cols_to_show = [
    "geoid", "year", "igs_total", "igs_place", "igs_economy", "igs_community",
    "NAME", "state", "county", "tract",
    "B19013_001E", "B19301_001E", "B25077_001E",
    "median_household_income", "per_capita_income", "median_home_value"
]
print(eda_df[cols_to_show].head())

```

EDA shape: (680256, 224)

EDA years: [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]

EDA unique tracts: 85032

Missing raw ACS columns: [] count = 0

Missing engineered feature columns: [] count = 0

Missing IGS columns: [] count = 0

Duplicate geoid-year rows: 0

	geoid	year	igs_total	igs_place	igs_economy	igs_community	
NAME	state	county	tract	B19013_001E	B19301_001E	B25077_001E	\
0	01001020100	2017	47.0	53.0	34.0	54.0	Cens
us Tract 201, Autauga County, Alabama				01	001	020100	67826.0
33018.0			152500.0				
1	01001020100	2018	52.0	48.0	48.0	62.0	Cens
us Tract 201, Autauga County, Alabama				01	001	020100	58625.0
31580.0			133300.0				
2	01001020100	2019	46.0	40.0	40.0	56.0	Cens
us Tract 201, Autauga County, Alabama				01	001	020100	60208.0
31225.0			136100.0				
3	01001020100	2020	46.0	34.0	48.0	56.0	Cens
us Tract 201, Autauga County, Alabama				01	001	020100	60388.0
30447.0			151200.0				
4	01001020100	2021	38.0	43.0	32.0	40.0	Cens
us Tract 201, Autauga County, Alabama				01	001	020100	57399.0
30934.0			138700.0				

	median_household_income	per_capita_income	median_home_value
0	67826.0	33018.0	152500.0
1	58625.0	31580.0	133300.0
2	60208.0	31225.0	136100.0
3	60388.0	30447.0	151200.0
4	57399.0	30934.0	138700.0

Cell 9 — Final checks before EDA

```
In [147... print("Merged years:", sorted(model_df_full["year"].dropna().unique().tolist))
print()

print(model_df_full[[
    "geoid", "year", "igs_total", "igs_place", "igs_economy", "igs_community",
    "affordable_housing_share", "internet_sub_share", "insured_share",
    "median_household_income", "poverty_rate", "vacancy_rate"
]].head())

print()
print("Top missingness in merged data:")
print(model_df_full.isna().mean().sort_values(ascending=False).head(30))

print()
print("Core numeric summary:")
core_cols = [
    "igs_total",
    "igs_place",
    "igs_economy",
    "igs_community",
    "affordable_housing_share",
    "internet_sub_share",
    "insured_share",
    "median_household_income",
    "per_capita_income",
    "poverty_rate",
    "vacancy_rate",
    "unemp_rate",
]
print(model_df_full[core_cols].describe())
```

Merged years: [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]

	geoid	year	igs_total	igs_place	igs_economy	igs_community	affordable_housing_share	internet_sub_share	insured_share	median_household_income	poverty_rate	vacancy_rate
0	01001020100	2017	47.0	53.0	34.0	54.0	0.737401	0.762599	0.909485	67826.0	0.106775	0.014379
1	01001020100	2018	52.0	48.0	48.0	62.0	0.704575	0.717647	0.907436	58625.0	0.113365	0.017972
2	01001020100	2019	46.0	40.0	40.0	56.0	0.715092	0.740480	0.902659	60208.0	0.166583	0.022069
3	01001020100	2020	46.0	34.0	48.0	56.0	0.738817	0.810967	0.903658	60388.0	0.136528	0.023944
4	01001020100	2021	38.0	43.0	32.0	40.0	0.743842	0.840722	0.909548	57399.0	0.153546	0.057276

Top missingness in merged data:

Is an Opportunity Zone	1.000000
Spending per Capita Tract, %	1.000000
Spend Growth Base, %	1.000000
is_opp_zone	1.000000
Spending per Capita Base, %	1.000000
Spend Growth Tract, %	1.000000
median_gross_rent_growth	0.276211
median_home_value_growth	0.256606
median_household_income_growth	0.241214
occupied_units_growth	0.236046
per_capita_income_growth	0.234822
Minority/Women Owned Businesses Tract, %	0.159347
Minority/Women Owned Businesses Base, %	0.159347
Minority/Women Owned Businesses Score	0.159347
Early Education Enrollment Tract, %	0.157891
Early Education Enrollment Score	0.157891
B25064_001E	0.147337
median_gross_rent	0.147337
B25077_001E	0.131784
median_home_value	0.131784
early_ed_enroll_share	0.124228
B19013_001E	0.116411
median_household_income	0.116411
gini	0.113279
B19083_001E	0.113279
affordable_housing_share	0.112346
internet_sub_share	0.112336
owner_share	0.112336
renter_share	0.112336
occupied_share	0.112055
dtype: float64	

Core numeric summary:

igs_total	igs_place	igs_economy	igs_community	affordable_housing_share
-----------	-----------	-------------	---------------	--------------------------

	e_housing_share	internet_sub_share	insured_share	median_household_income	
per_capita_income		poverty_rate \			
count	674227.000000	677454.000000	679363.000000	677817.000000	
	603832.000000	603839.000000	604557.000000	601067.000000	
	605097.000000	604202.000000			
mean	50.092197	49.971285	50.313671	50.150358	
	0.654425	0.847756	0.910665	73893.783232	3
	7332.819006	0.145232			
std	10.325957	12.498407	13.613307	13.857690	
	0.129589	0.115830	0.073407	37973.270136	2
	0408.309777	0.120057			
min	0.000000	0.000000	0.000000	0.000000	
	0.000000	0.000000	0.000000	2499.000000	
	0.000000	0.000000			
25%	44.000000	42.000000	42.000000	40.000000	
	0.575375	0.789683	0.879251	47967.000000	2
	4444.000000	0.059760			
50%	50.100000	50.000000	51.300000	50.000000	
	0.673483	0.873674	0.930211	65578.000000	3
	2765.000000	0.111340			
75%	58.000000	58.000000	60.000000	60.000000	
	0.750096	0.933931	0.962620	90962.000000	4
	4589.000000	0.195118			
max	84.000000	100.000000	94.000000	100.000000	
	1.000000	1.000000	1.000000	250001.000000	69
	1205.000000	1.000000			

	vacancy_rate	unemp_rate
count	604030.000000	604505.000000
mean	0.111957	0.060183
std	0.107473	0.049349
min	0.000000	0.000000
25%	0.042771	0.028459
50%	0.082713	0.048499
75%	0.145941	0.077549
max	1.000000	1.000000

Cell 10 — Optional: save an EDA subset for only chosen tracts

```
In [148... # Cell 10 – final EDA panel cleanup + save

eda_df = model_df_full.copy() # use model_df_full, not model_df

# drop junk unnamed / NaN column names from the original IGS export
junk_cols = [c for c in eda_df.columns if pd.isna(c) or str(c).startswith("")]
print("Dropping junk columns:", junk_cols)

eda_df = eda_df.drop(columns=junk_cols, errors="ignore")

# optional: reorder key columns to the front
front_cols = [
    c for c in [
        "geoid", "year", "Census Tract FIPS code", "County", "State",
        "igs_total", "igs_place", "igs_economy", "igs_community"
    ]
```

```

    ] if c in eda_df.columns
]
other_cols = [c for c in eda_df.columns if c not in front_cols]
eda_df = eda_df[front_cols + other_cols]

print("EDA panel shape:", eda_df.shape)

if SAVE_INTERMEDIATE:
    safe_save_parquet(eda_df, OUT_DIR / "eda_panel_clean.parquet")

eda_df.head()

```

Dropping junk columns: []
EDA panel shape: (680256, 224)

Out[148...

	geoid	year	Census Tract FIPS code	County	State	igs_total	igs_place	igs
0	01001020100	2017	01001020100	Autauga County	Alabama	47.0	53.0	
1	01001020100	2018	01001020100	Autauga County	Alabama	52.0	48.0	
2	01001020100	2019	01001020100	Autauga County	Alabama	46.0	40.0	
3	01001020100	2020	01001020100	Autauga County	Alabama	46.0	34.0	
4	01001020100	2021	01001020100	Autauga County	Alabama	38.0	43.0	

5 rows × 224 columns

Cell 11 — validation checks

In [149... *# Cell 11 – validation checks*

```
print("Duplicate geoid-year rows:", eda_df.duplicated(subset=["geoid", "year"]

required_acs_raw = ["NAME", "state", "county", "tract"] + ACS_VARS
missing_acs_raw = [c for c in required_acs_raw if c not in eda_df.columns]

required_feats = [c for c in feats.columns if c not in ["geoid", "year"]]
missing_feats = [c for c in required_feats if c not in eda_df.columns]

missing_igs = [c for c in igs.columns if c not in eda_df.columns]

print("Missing raw ACS columns:", len(missing_acs_raw))
print("Missing engineered ACS columns:", len(missing_feats))
print("Missing IGS columns:", len(missing_igs))
```

Duplicate geoid-year rows: 0
Missing raw ACS columns: 0
Missing engineered ACS columns: 0
Missing IGS columns: 0

Cell 12 — shortlist settings + required columns

In [150... *# Cell 12 – shortlist settings + required columns*

```
SHORTLIST_YEAR = 2024
LOW_IGS_THRESHOLD = 45
PERSIST_YEARS = [2022, 2023, 2024]

TOP_N_NATIONAL = 150
TOP_N_PER_STATE = 3
MIN_POP_FLAG = 1500 # flag only, do not drop automatically

shortlist_required_cols = [
    "geoid", "year",
    "NAME", "state", "county", "tract",
    "County", "State",
    "igs_total", "igs_place", "igs_economy", "igs_community",
    "pop_total", "share_under18", "share_65plus",
    "lfpr_16p", "unemp_rate",
    "median_household_income", "poverty_rate",
    "internet_sub_share", "vacancy_rate",
    "affordable_housing_share", "insured_share",
]

missing_shortlist_cols = [c for c in shortlist_required_cols if c not in eda
print("Missing shortlist columns:", missing_shortlist_cols)

if missing_shortlist_cols:
    raise ValueError(f"Missing required shortlist columns: {missing_shortlis

print("Shortlist year:", SHORTLIST_YEAR)
print("Low-IGS threshold:", LOW_IGS_THRESHOLD)
print("Persistence years:", PERSIST_YEARS)
```

Missing shortlist columns: []
Shortlist year: 2024
Low-IGS threshold: 45
Persistence years: [2022, 2023, 2024]

Cell 13 — scoring helpers

```
In [151]... # Cell 13 – scoring helpers

def weighted_percentile_points(series: pd.Series, weight: float, higher_is_worse: bool) -> float:
    """
    Convert a column into weighted percentile-rank points.
    - higher_is_worse=True -> larger values get more points
    - higher_is_worse=False -> smaller values get more points
    """
    ranked = series.rank(
        method="average",
        pct=True,
        ascending=higher_is_worse
    )
    return ranked * weight

def add_score(df: pd.DataFrame, col: str, weight: float, higher_is_worse: bool) -> pd.DataFrame:
    df[out_col] = weighted_percentile_points(df[col], weight=weight, higher_is_worse=higher_is_worse)
    return df
```

Cell 14 — build 2024 base + persistence table

```
In [152]... # Cell 14 – build 2024 base + persistence table

base_2024 = eda_df.loc[eda_df["year"] == SHORTLIST_YEAR].copy()

# display fields: prefer human-readable IGS fields, then fall back to ACS fields
base_2024["display_state"] = base_2024["State"].fillna(base_2024["state"])
base_2024["display_county"] = base_2024["County"].fillna(base_2024["county"])
base_2024["display_name"] = base_2024["NAME"]

persistence = (
    eda_df.loc[eda_df["year"].isin(PERSIST_YEARS), ["geoid", "year", "igs_total"]]
    .dropna(subset=["igs_total"])
    .assign(igs_below_45=lambda d: d["igs_total"] < LOW_IGS_THRESHOLD)
    .groupby("geoid", as_index=False)
    .agg(
        n_persist_years=("igs_below_45", "sum"),
        mean_igs_persist_window=("igs_total", "mean"),
        min_igs_persist_window=("igs_total", "min"),
    )
)

persistence["persistence_score"] = persistence["n_persist_years"].clip(lower=1, upper=3)

shortlist_base = (
    base_2024
    .merge(persistence, on="geoid", how="left")
)
```

```

        .copy()
    )

    shortlist_base["n_persist_years"] = shortlist_base["n_persist_years"].fillna(0)
    shortlist_base["persistence_score"] = shortlist_base["persistence_score"].fillna(0)

    # main eligibility rule
    shortlist_base = shortlist_base.loc[shortlist_base["igs_total"] < LOW_IGS_THRESHOLD]

    # tiny tract flag only
    shortlist_base["tiny_tract_flag"] = shortlist_base["pop_total"] < MIN_POP_FLAG

    print("2024 tracts total:", len(base_2024))
    print("Eligible low-IGS tracts:", len(shortlist_base))
    print("Unique states in shortlist:", shortlist_base["display_state"].nunique())
    print()

    print(shortlist_base[
        [
            "geoid", "display_state", "display_county", "display_name",
            "igs_total", "igs_place", "igs_economy", "igs_community",
            "n_persist_years", "persistence_score"
        ]
    ].head())

```

2024 tracts total: 85032
 Eligible low-IGS tracts: 24329
 Unique states in shortlist: 52

	geoid	display_state	display_county	display_name	igs_total	igs_place	igs_economy	igs_community	n_persist_years	persistence_score
0	01001020100	Alabama	Autauga County	Census Tract 201; Autauga County; Alabama	40.0	43.0	38.0	40.0	3	3.0
10	01001020803	Alabama	Autauga County	Census Tract 208.03; Autauga County; Alabama	42.0	30.0	56.0	39.0	1	1.0
11	01001020804	Alabama	Autauga County	Census Tract 208.04; Autauga County; Alabama	42.0	30.0	56.0	39.0	1	1.0
12	01001020805	Alabama	Autauga County	Census Tract 208.05; Autauga County; Alabama	42.0	30.0	56.0	39.0	1	1.0
13	01001020901	Alabama	Autauga County	Census Tract 209.01; Autauga County; Alabama	42.0	34.0	44.0	51.0	2	2.0

Cell 15 — apply the exact shortlist scorecard

In [153... *# Cell 15 – apply the exact shortlist scorecard*

```

score_input_cols = [
    "igs_total", "igs_place", "igs_economy", "igs_community",
    "poverty_rate", "insured_share", "median_household_income",

```

```

    "unemp_rate", "affordable_housing_share", "internet_sub_share",
    "share_65plus", "share_under18", "lfpr_16p", "vacancy_rate"
]

tract_shortlist_scored = shortlist_base.dropna(subset=score_input_cols).copy

# -----
# A. IGS distress (45 total)
# -----
tract_shortlist_scored = add_score(tract_shortlist_scored, "igs_total", 20,
tract_shortlist_scored = add_score(tract_shortlist_scored, "igs_economy", 10,
tract_shortlist_scored = add_score(tract_shortlist_scored, "igs_place", 8,
tract_shortlist_scored = add_score(tract_shortlist_scored, "igs_community",

tract_shortlist_scored["igs_distress_score"] = (
    tract_shortlist_scored["igs_total_pts"] +
    tract_shortlist_scored["igs_economy_pts"] +
    tract_shortlist_scored["igs_place_pts"] +
    tract_shortlist_scored["igs_community_pts"] +
    tract_shortlist_scored["persistence_score"]
)

# -----
# B. Healthcare proxy pressure (35 total)
# -----
tract_shortlist_scored = add_score(tract_shortlist_scored, "poverty_rate", 10,
tract_shortlist_scored = add_score(tract_shortlist_scored, "insured_share", 10,
tract_shortlist_scored = add_score(tract_shortlist_scored, "median_household_income", 10,
tract_shortlist_scored = add_score(tract_shortlist_scored, "unemp_rate", 5,
tract_shortlist_scored = add_score(tract_shortlist_scored, "affordable_housing_share", 5,
tract_shortlist_scored = add_score(tract_shortlist_scored, "internet_sub_share", 5,

tract_shortlist_scored["healthcare_proxy_score"] = (
    tract_shortlist_scored["poverty_rate_pts"] +
    tract_shortlist_scored["insured_share_pts"] +
    tract_shortlist_scored["median_household_income_pts"] +
    tract_shortlist_scored["unemp_rate_pts"] +
    tract_shortlist_scored["affordable_housing_share_pts"] +
    tract_shortlist_scored["internet_sub_share_pts"]
)

# -----
# C. Vulnerability / need (20 total)
# -----
tract_shortlist_scored = add_score(tract_shortlist_scored, "share_65plus", 6,
tract_shortlist_scored = add_score(tract_shortlist_scored, "share_under18", 6,
tract_shortlist_scored = add_score(tract_shortlist_scored, "lfpr_16p", 5,
tract_shortlist_scored = add_score(tract_shortlist_scored, "vacancy_rate", 5,

tract_shortlist_scored["vulnerability_score"] = (
    tract_shortlist_scored["share_65plus_pts"] +
    tract_shortlist_scored["share_under18_pts"] +
    tract_shortlist_scored["lfpr_16p_pts"] +
    tract_shortlist_scored["vacancy_rate_pts"]
)

```

```

# final total
tract_shortlist_scored["total_shortlist_score"] = (
    tract_shortlist_scored["igs_distress_score"] +
    tract_shortlist_scored["healthcare_proxy_score"] +
    tract_shortlist_scored["vulnerability_score"]
)

# keep a clean ranking
tract_shortlist_scored = tract_shortlist_scored.sort_values(
    ["total_shortlist_score", "igs_total"],
    ascending=[False, True]
).reset_index(drop=True)

print("Scored shortlist rows:", len(tract_shortlist_scored))
print()

print(tract_shortlist_scored[
    [
        "geoid", "display_state", "display_county", "display_name",
        "igs_total", "igs_place", "igs_economy", "igs_community",
        "igs_distress_score", "healthcare_proxy_score", "vulnerability_score",
        "total_shortlist_score"
    ]
].head(10))

```

Scored shortlist rows: 23873

	geoid	display_state	display_county	display_name	igs_total	igs_place	igs_economy	igs_community	igs_distress
0	47157011500	Tennessee	Shelby County	Census Tract 115; Sh					
		elby County; Tennessee	25.0	25.0	28.0	23.0			
	42.363716	32.687010							
1	72127003300	Puerto Rico	San Juan Municipio	Census Tract 33; San Juan					
		Municipio; Puerto Rico	22.0	32.0	21.0	14.0			
	42.547313	32.940812							
2	47157005900	Tennessee	Shelby County	Census Tract 59; Sh					
		elby County; Tennessee	22.0	28.0	30.0	8.0			
	41.972731	32.325514							
3	72127003600	Puerto Rico	San Juan Municipio	Census Tract 36; San Juan					
		Municipio; Puerto Rico	25.1	32.8	18.6	23.7			
	42.073933	33.985632							
4	04017940012	Arizona	Navajo County	Census Tract 9400.12;					
		Navajo County; Arizona	24.0	33.0	21.0	16.0			
	42.128932	31.208311							
5	47157011200	Tennessee	Shelby County	Census Tract 112; Sh					
		elby County; Tennessee	22.0	20.0	23.0	20.0			
	43.787752	31.509027							
6	46071941200	South Dakota	Jackson County	Census Tract 9412; Jackso					
		n County; South Dakota	18.0	20.0	22.0	8.0			
	44.121853	32.531123							
7	29510110100	Missouri	St. Louis city	Census Tract 1101; S					
		t. Louis city; Missouri	22.0	30.0	12.0	24.0			
	43.235454	33.879864							
8	04017942300	Arizona	Navajo County	Census Tract 9423;					
		Navajo County; Arizona	24.0	24.0	26.0	23.0			
	42.893562	30.229548							
9	12095010400	Florida	Orange County	Census Tract 104;					
		Orange County; Florida	24.0	24.0	22.0	26.0			
	43.250576	31.168768							

	vulnerability_score	total_shortlist_score
0	16.708373	91.759100
1	15.601391	91.089515
2	16.331630	90.629875
3	14.168726	90.228291
4	16.845348	90.182591
5	14.536380	89.833159
6	12.717296	89.370272
7	12.220123	89.335442
8	16.103422	89.226532
9	14.437942	88.857286

Cell 15.5 — shortlist audit

```
In [154... # Cell 15.5 – shortlist audit

print("=== SHORTLIST AUDIT ===")
print()
```



```

# 1) Row-count audit
eligible_n = len(shortlist_base)
scored_n = len(tract_shortlist_scored)
dropped_n = eligible_n - scored_n
dropped_pct = (dropped_n / eligible_n * 100) if eligible_n else 0

print("Row-count audit")
print(f"Eligible low-IGS tracts in shortlist_base: {eligible_n:,}")
print(f"Scored tracts in tract_shortlist_scored: {scored_n:,}")
print(f"Dropped before scoring: {dropped_n:,} ({dropped_pct:.2f}%)")
print()

# 2) Missingness audit for the scoring inputs
print("Missingness audit on score_input_cols within shortlist_base")
missing_summary = pd.DataFrame({
    "missing_count": shortlist_base[score_input_cols].isna().sum(),
    "missing_pct": shortlist_base[score_input_cols].isna().mean().mul(100)
}).sort_values(["missing_count", "missing_pct"], ascending=[False, False])

print(missing_summary)
print()

# 3) Which columns are causing drops specifically?
dropped_rows = shortlist_base[shortlist_base[score_input_cols].isna().any(axis=1)]

print("Rows dropped because at least one score input is missing:", len(dropped_rows))
print()

if len(dropped_rows) > 0:
    dropped_missing_summary = pd.DataFrame({
        "missing_count_among_dropped": dropped_rows[score_input_cols].isna().sum(),
        "missing_pct_among_dropped": dropped_rows[score_input_cols].isna().mean().mul(100)
    }).sort_values(["missing_count_among_dropped", "missing_pct_among_dropped"], ascending=[False, False])

    print("Missingness among dropped rows only")
    print(dropped_missing_summary)
    print()

    print("Sample dropped rows")
    display_cols = [
        "geoid", "display_state", "display_county", "display_name",
        "igs_total", "igs_place", "igs_economy", "igs_community"
    ] + score_input_cols

    print(dropped_rows[display_cols].head(10))
    print()

# 4) Score-construction audit
print("Score construction audit")

tract_shortlist_scored["igs_distress_score_check"] = (
    tract_shortlist_scored["igs_total_pts"] +
    tract_shortlist_scored["igs_economy_pts"] +
    tract_shortlist_scored["igs_place_pts"] +
    tract_shortlist_scored["igs_community_pts"] +
    tract_shortlist_scored["persistence_score"]
)

```

```

)

tract_shortlist_scored["healthcare_proxy_score_check"] = (
    tract_shortlist_scored["poverty_rate_pts"] +
    tract_shortlist_scored["insured_share_pts"] +
    tract_shortlist_scored["median_household_income_pts"] +
    tract_shortlist_scored["unemp_rate_pts"] +
    tract_shortlist_scored["affordable_housing_share_pts"] +
    tract_shortlist_scored["internet_sub_share_pts"]
)

tract_shortlist_scored["vulnerability_score_check"] = (
    tract_shortlist_scored["share_65plus_pts"] +
    tract_shortlist_scored["share_under18_pts"] +
    tract_shortlist_scored["lfpr_16p_pts"] +
    tract_shortlist_scored["vacancy_rate_pts"]
)

tract_shortlist_scored["total_shortlist_score_check"] = (
    tract_shortlist_scored["igs_distress_score_check"] +
    tract_shortlist_scored["healthcare_proxy_score_check"] +
    tract_shortlist_scored["vulnerability_score_check"]
)

score_audit = pd.DataFrame({
    "igs_distress_max_abs_diff": [
        (tract_shortlist_scored["igs_distress_score"] - tract_shortlist_scored["igs_distress_score"].min()),
    ],
    "healthcare_proxy_max_abs_diff": [
        (tract_shortlist_scored["healthcare_proxy_score"] - tract_shortlist_scored["healthcare_proxy_score"].min()),
    ],
    "vulnerability_max_abs_diff": [
        (tract_shortlist_scored["vulnerability_score"] - tract_shortlist_scored["vulnerability_score"].min()),
    ],
    "total_score_max_abs_diff": [
        (tract_shortlist_scored["total_shortlist_score"] - tract_shortlist_scored["total_shortlist_score"].min()),
    ],
})

print(score_audit)
print()

# 5) Score-range audit
print("Score range audit")
range_audit = pd.DataFrame({
    "min": tract_shortlist_scored[
        ["igs_distress_score", "healthcare_proxy_score", "vulnerability_score"]
    ].min(),
    "max": tract_shortlist_scored[
        ["igs_distress_score", "healthcare_proxy_score", "vulnerability_score"]
    ].max(),
    "mean": tract_shortlist_scored[
        ["igs_distress_score", "healthcare_proxy_score", "vulnerability_score"]
    ].mean(),
})
print(range_audit)

```

```

print()

# 6) Persistence audit
print("Persistence audit")
print(tract_shortlist_scored["n_persist_years"].value_counts(dropna=False).s
print()
print("Persistence score min/max:",
      tract_shortlist_scored["persistence_score"].min(),
      tract_shortlist_scored["persistence_score"].max())
print()

# 7) Tiny-tract audit among top-ranked rows
print("Tiny-tract audit in top 25 scored rows")
print(
    tract_shortlist_scored.head(25)[
        ["geoid", "display_state", "display_county", "igs_total", "total_sho
    ]
)
print()

print("=== END SHORTLIST AUDIT ===")

```

=== SHORTLIST AUDIT ===

Row-count audit

Eligible low-IGS tracts in shortlist_base: 24,329

Scored tracts in tract_shortlist_scored: 23,873

Dropped before scoring: 456 (1.87%)

Missingness audit on score_input_cols within shortlist_base

	missing_count	missing_pct
median_household_income	456	1.874306
affordable_housing_share	227	0.933043
internet_sub_share	227	0.933043
vacancy_rate	220	0.904271
poverty_rate	213	0.875498
insured_share	197	0.809733
unemp_rate	192	0.789182
share_65plus	143	0.587776
share_under18	143	0.587776
lfpr_16p	143	0.587776
igs_total	0	0.000000
igs_place	0	0.000000
igs_economy	0	0.000000
igs_community	0	0.000000

Rows dropped because at least one score input is missing: 456

Missingness among dropped rows only

	missing_count_among_dropped	missing_pct_among_dro
pped		
median_household_income	456	100.00
0000		
affordable_housing_share	227	49.78
0702		
internet_sub_share	227	49.78
0702		
vacancy_rate	220	48.24
5614		
poverty_rate	213	46.71
0526		
insured_share	197	43.20
1754		
unemp_rate	192	42.10
5263		
share_65plus	143	31.35
9649		
share_under18	143	31.35
9649		
lfpr_16p	143	31.35
9649		
igs_total	0	0.00
0000		
igs_place	0	0.00
0000		
igs_economy	0	0.00
0000		
igs_community	0	0.00

0000

Sample dropped rows

	geoid	display_state	display_county				
display_name	igs_total	igs_place	igs_economy	igs_community	igs_total	i	
gs_place	igs_economy	\					
141	01015981901	Alabama	Calhoun County	Census Tract	9819.01;		
Calhoun County; Alabama	0.0	0.0	5.0	0.0			
0.0	0.0	5.0					
142	01015981902	Alabama	Calhoun County	Census Tract	9819.02;		
Calhoun County; Alabama	0.1	0.1	0.1	0.1			
0.1	0.1	0.1					
418	01065040200	Alabama	Hale County	Census Tract	4		
02; Hale County; Alabama	39.0	33.0	36.0	48.0			
39.0	33.0	36.0					
543	01073010605	Alabama	Jefferson County	Census Tract	106.05; J		
efferson County; Alabama	44.0	48.0	44.0	38.0			
44.0	48.0	44.0					
981	01097003602	Alabama	Mobile County	Census Tract	36.0		
2; Mobile County; Alabama	29.0	15.0	40.0	32.0			
29.0	15.0	40.0					
1211	01113030904	Alabama	Russell County	Census Tract	309.04;		
Russell County; Alabama	40.0	36.0	41.0	44.0			
40.0	36.0	41.0					
1295	01117980000	Alabama	Shelby County	Census Tract	980		
0; Shelby County; Alabama	0.2	0.2	40.0	0.1			
0.2	0.2	40.0					
1593	02185000300	Alaska	North Slope Borough	Census Tract	3; Nor		
th Slope Borough; Alaska	0.0	0.0	63.0	68.0			
0.0	0.0	63.0					
1704	04005980000	Arizona	Coconino County	Census Tract	9800;		
Coconino County; Arizona	35.9	47.2	61.7	28.0			
35.9	47.2	61.7					
1807	04013061017	Arizona	Maricopa County	Census Tract	610.17;		
Maricopa County; Arizona	2.1	1.3	2.6	2.3			
2.1	1.3	2.6					

	igs_community	poverty_rate	insured_share	median_household_income	u		
nemp_rate	affordable_housing_share	internet_sub_share	share_65plus	share			
_under18	lfpr_16p	vacancy_rate					
141	0.0	NaN	NaN	NaN			
NaN		NaN	NaN	NaN			N
aN	NaN	NaN					
142	0.1	NaN	NaN	NaN			
NaN		NaN	NaN	NaN			N
aN	NaN	NaN					
418	48.0	0.168217	1.000000	NaN			
0.054575		0.771894	0.720978	0.185271			
0.142636	0.537532	0.285298					
543	38.0	0.589583	0.766667	NaN			
0.136126		0.308880	0.706564	0.045833			
0.356250	0.565089	0.457023					
981	32.0	0.285271	0.839705	NaN			
0.015534		0.504425	0.865044	0.085430			
0.338411	0.515516	0.178182					
1211	44.0	0.146334	0.919027	NaN			

0.106940		0.749132	0.800347	0.213156	
0.267552	0.436242	0.012853			
1295	0.1	NaN	NaN		NaN
NaN		NaN	NaN	NaN	N
aN	NaN	NaN			
1593	68.0	0.041571	0.963768		NaN
0.010510		NaN	NaN	0.030038	
0.000000	0.979848	NaN			
1704	28.0	0.465839	0.894410		NaN
0.000000		1.000000	1.000000	0.000000	
0.000000	0.708075	0.880952			
1807	2.3	NaN	NaN		NaN
NaN		NaN	NaN	0.050380	0.0000
00	0.000000	NaN			

Score construction audit

igs_distress_max_abs_diff	healthcare_proxy_max_abs_diff	vulnerability_max_abs_diff	total_score_max_abs_diff
0	0.0	0.0	
0.0	0.0		

Score range audit

	min	max	mean
igs_distress_score	5.910108	44.925062	23.607758
healthcare_proxy_score	0.202153	34.448038	17.500733
vulnerability_score	0.322289	19.203493	10.000419
total_shortlist_score	15.470657	91.759100	51.108910

Persistence audit

```
n_persist_years
1      2160
2      5065
3     16648
Name: count, dtype: int64
```

Persistence score min/max: 1.0 3.0

Tiny-tract audit in top 25 scored rows

st_score	geoid	display_state	display_county	igs_total	total_shortli
0	47157011500	Tennessee	Shelby County	25.0	9
1.759100		False			
1	72127003300	Puerto Rico	San Juan Municipio	22.0	9
1.089515		True			
2	47157005900	Tennessee	Shelby County	22.0	9
0.629875		False			
3	72127003600	Puerto Rico	San Juan Municipio	25.1	9
0.228291		True			
4	04017940012	Arizona	Navajo County	24.0	9
0.182591		False			
5	47157011200	Tennessee	Shelby County	22.0	8
9.833159		True			
6	46071941200	South Dakota	Jackson County	18.0	8
9.370272		False			
7	29510110100	Missouri	St. Louis city	22.0	8
9.335442		False			

8	04017942300	Arizona	Navajo County	24.0	8
9	226532	False			
9	12095010400	Florida	Orange County	24.0	8
	8.857286	True			
10	22019000200	Louisiana	Calcasieu Parish	24.0	8
	8.815189	True			
11	54069000700	West Virginia	Ohio County	23.0	8
	8.797177	True			
12	29510107200	Missouri	St. Louis city	16.0	8
	8.718050	True			
13	18089011400	Indiana	Lake County	22.0	8
	8.707871	True			
14	01097000500	Alabama	Mobile County	24.0	8
	8.467327	True			
15	17031681200	Illinois	Cook County	19.0	8
	8.465442	False			
16	04001944301	Arizona	Apache County	21.0	8
	8.431911	False			
17	28059041600	Mississippi	Jackson County	26.0	8
	8.357894	False			
18	18089012100	Indiana	Lake County	25.0	8
	8.340908	True			
19	17031530100	Illinois	Cook County	20.0	8
	8.031018	False			
20	72013302201	Puerto Rico	Arecibo Municipio	24.0	8
	7.806225	False			
21	01097001502	Alabama	Mobile County	20.0	8
	7.668349	True			
22	01097001501	Alabama	Mobile County	18.0	8
	7.657374	True			
23	72127001400	Puerto Rico	San Juan Municipio	21.0	8
	7.593411	True			
24	01097004800	Alabama	Mobile County	16.0	8
	7.440875	True			

=== END SHORTLIST AUDIT ===

Cell 16 — build the main shortlist outputs

```
In [155... # Cell 16 – build the main shortlist outputs

shortlist_output_cols = [
    "geoid", "year",
    "display_state", "display_county", "display_name",
    "State", "County", "state", "county", "tract",
    "igs_total", "igs_place", "igs_economy", "igs_community",
    "n_persist_years", "mean_igs_persist_window", "min_igs_persist_window",
    "pop_total", "share_under18", "share_65plus",
    "lfpr_16p", "unemp_rate", "median_household_income", "poverty_rate",
    "internet_sub_share", "vacancy_rate", "affordable_housing_share", "insur
    "tiny_tract_flag",
    "igs_distress_score", "healthcare_proxy_score", "vulnerability_score",
    "total_shortlist_score"
]
```

```

tract_shortlist_scored = tract_shortlist_scored[shortlist_output_cols].copy()

top_national_candidates = tract_shortlist_scored.head(TOP_N_NATIONAL).copy()

state_balanced_candidates = (
    tract_shortlist_scored
    .sort_values(["display_state", "total_shortlist_score"], ascending=[True, False])
    .groupby("display_state", group_keys=False)
    .head(TOP_N_PER_STATE)
    .reset_index(drop=True)
)

print("Top national candidates:", top_national_candidates.shape)
print("State-balanced candidates:", state_balanced_candidates.shape)
print()

print(top_national_candidates[
    ["geoid", "display_state", "display_county", "igs_total", "total_shortli
]].head(20))

print()
print(state_balanced_candidates[
    ["display_state", "geoid", "display_county", "igs_total", "total_shortli
]].head(30))

```


Top national candidates: (150, 33)
 State-balanced candidates: (153, 33)

	geoid	display_state	display_county	igs_total	total_shortli
st_score					
0	47157011500	Tennessee	Shelby County	25.0	9
1.759100					
1	72127003300	Puerto Rico	San Juan Municipio	22.0	9
1.089515					
2	47157005900	Tennessee	Shelby County	22.0	9
0.629875					
3	72127003600	Puerto Rico	San Juan Municipio	25.1	9
0.228291					
4	04017940012	Arizona	Navajo County	24.0	9
0.182591					
5	47157011200	Tennessee	Shelby County	22.0	8
9.833159					
6	46071941200	South Dakota	Jackson County	18.0	8
9.370272					
7	29510110100	Missouri	St. Louis city	22.0	8
9.335442					
8	04017942300	Arizona	Navajo County	24.0	8
9.226532					
9	12095010400	Florida	Orange County	24.0	8
8.857286					
10	22019000200	Louisiana	Calcasieu Parish	24.0	8
8.815189					
11	54069000700	West Virginia	Ohio County	23.0	8
8.797177					
12	29510107200	Missouri	St. Louis city	16.0	8
8.718050					
13	18089011400	Indiana	Lake County	22.0	8
8.707871					
14	01097000500	Alabama	Mobile County	24.0	8
8.467327					
15	17031681200	Illinois	Cook County	19.0	8
8.465442					
16	04001944301	Arizona	Apache County	21.0	8
8.431911					
17	28059041600	Mississippi	Jackson County	26.0	8
8.357894					
18	18089012100	Indiana	Lake County	25.0	8
8.340908					
19	17031530100	Illinois	Cook County	20.0	8
8.031018					

	display_state	geoid	display_county	igs_total
total_shortlist_score				
0	Alabama	01097000500	Mobile County	24.0
88.467327				
1	Alabama	01097001502	Mobile County	20.0
87.668349				
2	Alabama	01097001501	Mobile County	18.0
87.657374				
3	Alaska	02290000100	Yukon-Koyukuk Census Area	32.0
72.989570				

4 67.722511	Alaska	02170000401	Matanuska-Susitna Borough	30.0
5 65.590144	Alaska	02050000300	Bethel Census Area	36.0
6 90.182591	Arizona	04017940012	Navajo County	24.0
7 89.226532	Arizona	04017942300	Navajo County	24.0
8 88.431911	Arizona	04001944301	Apache County	21.0
9 84.442131	Arkansas	05077470100	Lee County	22.1
10 82.394839	Arkansas	05017080200	Chicot County	28.0
11 81.988858	Arkansas	05035031100	Crittenden County	24.0
12 81.865999	California	06025012400	Imperial County	28.0
13 80.098877	California	06037209403	Los Angeles County	32.0
14 79.349726	California	06037530902	Los Angeles County	28.0
15 76.857286	Colorado	08123000600	Weld County	30.0
16 73.474511	Colorado	08123000501	Weld County	32.0
17 68.479412	Colorado	08123001906	Weld County	26.0
18 65.779123	Delaware	10003002200	New Castle County	38.0
19 64.864994	Delaware	10003003002	New Castle County	39.8
20 64.245005	Delaware	10005050703	Sussex County	34.0
21 76.194006	District of Columbia	11001007406	District of Columbia	28.0
22 75.363653	District of Columbia	11001007401	District of Columbia	30.2
23 74.402254	District of Columbia	11001009601	District of Columbia	28.0
24 88.857286	Florida	12095010400	Orange County	24.0
25 87.210824	Florida	12095010500	Orange County	26.0
26 85.219055	Florida	12086002001	Miami-Dade County	29.0
27 87.424350	Georgia	13193000200	Macon County	12.0
28 86.301575	Georgia	13275960100	Thomas County	26.0
29 85.505802	Georgia	13299950400	Ware County	30.0

Cell 17 — county-level candidate community seeds

In [156... *# Cell 17 – county-level candidate community seeds*

```
candidate_communities = (  
    top_national_candidates  
    .groupby(["display_state", "display_county"], dropna=False)  
    .agg(  
        n_shortlisted_tracts=("geoid", "nunique"),  
        mean_shortlist_score=("total_shortlist_score", "mean"),  
        max_shortlist_score=("total_shortlist_score", "max"),  
        mean_igs_total=("igs_total", "mean"),  
        mean_poverty_rate=("poverty_rate", "mean"),  
        mean_insured_share=("insured_share", "mean"),  
        mean_unemp_rate=("unemp_rate", "mean"),  
        mean_affordable_housing_share=("affordable_housing_share", "mean"),  
        total_pop=("pop_total", "sum"),  
        sample_tracts=("geoid", lambda s: ", ".join(s.astype(str).head(5))),  
        sample_names=("display_name", lambda s: " | ".join(s.astype(str).head(5)))  
    )  
    .reset_index()  
    .sort_values(  
        ["n_shortlisted_tracts", "mean_shortlist_score", "max_shortlist_score"],  
        ascending=[False, False, False]  
    )  
    .reset_index(drop=True)  
)  
  
print(candidate_communities.head(25))
```

	display_state	display_county	n_shortlisted_tracts	mean_shortlist_score	max_shortlist_score	mean_igs_total	mean_poverty_rate	mean_insured_share	mean_unemp_rate	\
0	Puerto Rico	San Juan Municipio	14	8	5.792400	91.089515	25.614286	0.607932	0.819181	0.203197
1	Illinois	Cook County	9	8	6.357391	88.465442	22.666667	0.397891	0.849558	0.235932
2	Alabama	Mobile County	8	8	6.916284	88.467327	22.000000	0.438018	0.824336	0.196453
3	Tennessee	Shelby County	7	8	7.296375	91.759100	22.857143	0.478255	0.823091	0.241875
4	Ohio	Cuyahoga County	5	8	6.015930	87.098270	22.520000	0.435460	0.840377	0.122584
5	Michigan	Wayne County	5	8	5.224530	86.572949	21.840000	0.518372	0.911347	0.199342
6	Louisiana	Orleans Parish	5	8	4.754338	87.221820	25.600000	0.430523	0.780681	0.200345
7	Georgia	Bibb County	5	8	4.695112	85.227160	27.400000	0.500733	0.843430	0.201833
8	Indiana	Lake County	4	8	7.046695	88.707871	24.250000	0.491659	0.868145	0.169811
9	Puerto Rico	Arecibo Municipio	4	8	5.504765	87.806225	21.750000	0.645395	0.946609	0.214093
10	Arizona	Navajo County	3	8	8.642874	90.182591	16.000000	0.432138	0.836338	0.164271
11	Missouri	St. Louis city	3	8	7.338988	89.335442	20.000000	0.446765	0.779344	0.250193
12	Arizona	Apache County	3	8	6.701615	88.431911	22.666667	0.414109	0.792306	0.150025
13	Puerto Rico	Guánica Municipio	3	8	4.968318	85.605203	22.666667	0.664305	0.949924	0.265815
14	Florida	Orange County	2	8	8.034055	88.857286	25.000000	0.520890	0.712983	0.160491
15	West Virginia	Ohio County	2	8	7.012158	88.797177	24.800000	0.416770	0.838375	0.183604
16	New Mexico	San Juan County	2	8	5.924402	87.008022	13.000000	0.425332	0.798328	0.120984
17	Alabama	Montgomery County	2	8	4.161982	84.756545	27.000000	0.376847		

0.794372	0.123590		
18	Puerto Rico	Río Grande Municipio	2 8
4.005152	84.659867	22.600000	0.461223
0.892799	0.180524		
19	Texas	Jefferson County	2 8
4.004566	84.166757	27.500000	0.425718
0.775777	0.120761		
20	South Dakota	Jackson County	1 8
9.370272	89.370272	18.000000	0.513158
0.672216	0.117308		
21	Louisiana	Calcasieu Parish	1 8
8.815189	88.815189	24.000000	0.611966
0.895093	0.273810		
22	Mississippi	Jackson County	1 8
8.357894	88.357894	26.000000	0.461949
0.704314	0.132479		
23	Georgia	Macon County	1 8
7.424350	87.424350	12.000000	0.370725
0.836525	0.084034		
24	Texas	Bowie County	1 8
7.404704	87.404704	26.000000	0.540254
0.757384	0.130769		

	mean_affordable_housing_share	total_pop	
sample_tracts \			
0	0.455896	26214.0	72127003300, 72127003600, 72127001400, 72127003701, 72127002800
1	0.343617	13497.0	17031681200, 17031530100, 17031691500, 17031826902, 17031681300
2	0.414796	9149.0	01097000500, 01097001502, 01097001501, 01097004800, 01097001302
3	0.447223	12627.0	47157011500, 47157005900, 47157011200, 47157000600, 47157006000
4	0.391700	6914.0	39035119900, 39035116300, 39035150100, 39035151500, 39035151600
5	0.394520	8033.0	26163530900, 26163530800, 26163515200, 26163533600, 26163514300
6	0.393170	11449.0	22071000903, 22071001401, 22071001746, 22071002700, 22071007501
7	0.444068	11293.0	13021011500, 13021012400, 13021012900, 13021012800, 13021012700
8	0.315927	3030.0	18089011400, 18089012100, 18089011900, 18089010600
9	0.432804	7691.0	72013302201, 72013301600, 72013300500, 72013302300
10	0.646109	6353.0	04017940014, 04017942300, 04017940012, 04017942300
11	0.388020	4436.0	29510107200, 29510107400, 2951010100, 29510107200
12	0.689441	11755.0	04001944100, 04001944302, 04001944301
13	0.593534	4827.0	72055961200, 72055961300, 72055961500
14	0.174092	2080.0	12095010400, 12095010500
15	0.421457	1592.0	

54069000700, 54069000400		
16	0.677725	5707.0
35045943100, 35045943000		
17	0.452788	5080.0
01101003000, 01101002201		
18	0.608788	6595.0
72119130601, 72119130402		
19	0.424027	2992.0
48245005900, 48245000900		
20	0.495885	1922.0
46071941200		
21	0.450667	1182.0
22019000200		
22	0.431818	2063.0
28059041600		
23	0.504673	1547.0
13193000200		
24	0.296044	1896.0
48037010600		

sample_names

0	Census Tract 33; San Juan Municipio; Puerto Rico Census Tract 36; San Juan Municipio; Puerto Rico Census Tract 1...
1	Census Tract 6812; Cook County; Illinois Census Tract 5301; Cook County; Illinois Census Tract 6915; Cook County...
2	Census Tract 5; Mobile County; Alabama Census Tract 15.02; Mobile County; Alabama Census Tract 15.01; Mobile Cou...
3	Census Tract 115; Shelby County; Tennessee Census Tract 59; Shelby County; Tennessee Census Tract 112; Shelby Co...
4	Census Tract 1199; Cuyahoga County; Ohio Census Tract 1163; Cuyahoga County; Ohio Census Tract 1501; Cuyahoga Co...
5	Census Tract 5309; Wayne County; Michigan Census Tract 5308; Wayne County; Michigan Census Tract 5152; Wayne Cou...
6	Census Tract 9.03; Orleans Parish; Louisiana Census Tract 14.01; Orleans Parish; Louisiana Census Tract 17.46; O...
7	Census Tract 115; Bibb County; Georgia Census Tract 124; Bibb County; Georgia Census Tract 129; Bibb County; Geo...
8	Census Tract 114; Lake County; Indiana Census Tract 121; Lake County; Indiana Census Tract 119; Lake County; Ind...
9	Census Tract 3022.01; Arecibo Municipio; Puerto Rico Census Tract 3016; Arecibo Municipio; Puerto Rico Census Tr...
10	Census Tract 9400.12; Navajo County; Arizona Census Tract 9423; Navajo County; Arizona Census Tract 9400.14; Nav...
11	Census Tract 1101; St. Louis city; Missouri Census Tract 1072; St. Louis city; Missouri Census Tract 1074; St. L...
12	Census Tract 9443.01; Apache County; Arizona Census Tract 9443.02; Apache County; Arizona Census Tract 9441; Apa...
13	Census Tract 9615; Guánica Municipio; Puerto Rico Census Tract 9612; Guánica Municipio; Puerto Rico Census Tract...
14	Census Tract 104; Orange County; Florida Census Tract 105; Orange County; Florida
15	Census Tract 7; Ohio County; West Virginia Census Tract 4; Ohio County; West Virginia
16	Census Tract 9431; San Juan County; New Mexico Census Tract 9430; San Juan County; New Mexico

17 Census Tract 30; Montgomery County; Alabama |
 Census Tract 22.01; Montgomery County; Alabama
 18 Census Tract 1306.01; Río Grande Municipio; Puerto Rico | Census T
 ract 1304.02; Río Grande Municipio; Puerto Rico
 19 Census Tract 59; Jefferson County;
 Texas | Census Tract 9; Jefferson County; Texas
 20
 Census Tract 9412; Jackson County; South Dakota
 21
 Census Tract 2; Calcasieu Parish; Louisiana
 22
 Census Tract 416; Jackson County; Mississippi
 23
 Census Tract 2; Macon County; Georgia
 24
 Census Tract 106; Bowie County; Texas

Cell 18 — save outputs for team discussion

```
In [157... # Cell 18 – save shortlist outputs for team discussion

if SAVE_INTERMEDIATE:
    safe_save_parquet(tract_shortlist_scored, OUT_DIR / "tract_shortlist_sco
    safe_save_parquet(top_national_candidates, OUT_DIR / "top_national_candi
    safe_save_parquet(state_balanced_candidates, OUT_DIR / "state_balanced_c
    safe_save_parquet(candidate_communities, OUT_DIR / "candidate_communitie

tract_shortlist_scored.to_csv(OUT_DIR / "tract_shortlist_scored.csv", index=
top_national_candidates.to_csv(OUT_DIR / "top_national_candidates.csv", inde
state_balanced_candidates.to_csv(OUT_DIR / "state_balanced_candidates.csv",
candidate_communities.to_csv(OUT_DIR / "candidate_communities.csv", index=Fa

print("Saved shortlist outputs to:", OUT_DIR)
```

Saved shortlist outputs to: data\processed

Cell 19 — quick sanity checks before you present results to the team

```
In [158... # Cell 19 – quick sanity checks before sharing with the team

print("Top 10 states by count in national shortlist:")
print(top_national_candidates["display_state"].value_counts().head(10))

print()
print("Top 10 counties by count in national shortlist:")
print(
    top_national_candidates
    .assign(state_county=lambda d: d["display_county"].astype(str) + ", " +
    ["state_county"])
    .value_counts()
    .head(10)
)
```

```
print()
print("Tiny tract flags in national shortlist:")
print(top_national_candidates["tiny_tract_flag"].value_counts(dropna=False))

print()
print("Summary of final shortlist scores:")
print(top_national_candidates[
    ["igs_distress_score", "healthcare_proxy_score", "vulnerability_score",
    ].describe())
```


Top 10 states by count in national shortlist:

display_state

Puerto Rico	37
Alabama	16
Georgia	11
Illinois	10
Louisiana	9
Tennessee	8
Mississippi	8
Ohio	6
Arizona	6
Michigan	5

Name: count, dtype: int64

Top 10 counties by count in national shortlist:

state_county

San Juan Municipio, Puerto Rico	14
Cook County, Illinois	9
Mobile County, Alabama	8
Shelby County, Tennessee	7
Bibb County, Georgia	5
Wayne County, Michigan	5
Cuyahoga County, Ohio	5
Orleans Parish, Louisiana	5
Arecibo Municipio, Puerto Rico	4
Lake County, Indiana	4

Name: count, dtype: int64

Tiny tract flags in national shortlist:

tiny_tract_flag

False	99
True	51

Name: count, dtype: int64

Summary of final shortlist scores:

	igs_distress_score	healthcare_proxy_score	vulnerability_score	total_score
count	150.000000	150.000000	150.000000	
mean	41.646020	29.483917	14.576241	
std	1.978924	2.407299	1.614854	
min	35.852218	23.188539	10.070163	
25%	40.322865	28.065786	13.509383	
50%	42.023018	29.484491	14.672894	
75%	43.126932	31.468448	15.606292	
max	44.803586	34.020421	19.203493	

Cell 19.5 — broader adjacency scope

In [169... *# Cell 19.5 – broader adjacency scope*

```
SEED_COUNTIES_N = 15
BROADER_SCORE_FLOOR = 80.0
BROADER_MAX_PER_COUNTY = 20

# take the top counties that already surfaced in candidate_communities
seed_counties = (
    candidate_communities
    .head(SEED_COUNTIES_N)[["display_state", "display_county"]]
    .drop_duplicates()
    .copy()
)

# pull in more shortlisted tracts from those counties, not just the top-150
broader_county_pool = (
    tract_shortlist_scored
    .merge(seed_counties, on=["display_state", "display_county"], how="inner")
    .loc[lambda d: d["total_shortlist_score"] >= BROADER_SCORE_FLOOR]
    .copy()
)

# keep the strongest tracts per county so adjacency stays computationally manageable
broader_county_pool["county_rank_by_score"] = (
    broader_county_pool
    .groupby(["display_state", "display_county"])["total_shortlist_score"]
    .rank(method="first", ascending=False)
)

broader_county_pool = (
    broader_county_pool
    .loc[broader_county_pool["county_rank_by_score"] <= BROADER_MAX_PER_COUNTY]
    .drop(columns=["county_rank_by_score"])
    .copy()
)

# build the broader adjacency scope
CLUSTER_SCOPE_DF = (
    pd.concat(
        [
            top_national_candidates,
            broader_county_pool,
        ],
        ignore_index=True
    )
    .drop_duplicates(subset=["geoid"])
    .sort_values(
        ["display_state", "display_county", "total_shortlist_score"],
        ascending=[True, True, False]
    )
    .reset_index(drop=True)
)

print("Seed counties:", len(seed_counties))
print(seed_counties)
```

```

print()
print("Broader county pool shape:", broader_county_pool.shape)
print("Broader adjacency scope shape:", CLUSTER_SCOPE_DF.shape)
print("Unique counties in broader scope:",
      CLUSTER_SCOPE_DF[["display_state", "display_county"]].drop_duplicates()

print()
print("Top counties by tract count in broader scope:")
print(
    CLUSTER_SCOPE_DF
    .assign(state_county=lambda d: d["display_county"].astype(str) + ", " +
    ["state_county"]
    .value_counts()
    .head(20)
)

```

```
Seed counties: 15
  display_state      display_county
0    Puerto Rico  San Juan Municipio
1      Illinois      Cook County
2      Alabama      Mobile County
3    Tennessee      Shelby County
4        Ohio      Cuyahoga County
5      Michigan      Wayne County
6    Louisiana      Orleans Parish
7      Georgia      Bibb County
8      Indiana      Lake County
9    Puerto Rico  Arecibo Municipio
10     Arizona      Navajo County
11    Missouri      St. Louis city
12     Arizona      Apache County
13    Puerto Rico  Guánica Municipio
14     Florida      Orange County
```

```
Broader county pool shape: (143, 33)
Broader adjacency scope shape: (213, 33)
Unique counties in broader scope: 80
```

```
Top counties by tract count in broader scope:
```

```
state_county
San Juan Municipio, Puerto Rico      20
Cook County, Illinois                20
Wayne County, Michigan               16
Shelby County, Tennessee            14
Cuyahoga County, Ohio               11
Lake County, Indiana                10
Mobile County, Alabama               9
Orleans Parish, Louisiana            8
Arecibo Municipio, Puerto Rico       7
Apache County, Arizona               7
Bibb County, Georgia                 7
St. Louis city, Missouri             5
Guánica Municipio, Puerto Rico       4
Navajo County, Arizona               3
Orange County, Florida               2
Montgomery County, Alabama           2
San Juan County, New Mexico           2
Río Grande Municipio, Puerto Rico    2
Ohio County, West Virginia           2
Jefferson County, Texas              2
Name: count, dtype: int64
```

Cell 20 — adjacency settings + tract file setup

```
In [170... # Cell 20 – state-based tract download setup
```

```
import zipfile
import requests
from pathlib import Path
import pandas as pd
```

```

ADJACENCY_MODE = "shared_edge_or_corner"
# CLUSTER_SCOPE_DF = top_national_candidates.copy()
assert "CLUSTER_SCOPE_DF" in globals(), "Run Cell 19.5 before Cell 20"

MIN_CLUSTER_TRACTS = 3
MIN_CLUSTER_POP = 5000

TRACT_BASE_URL = "https://www2.census.gov/geo/tiger/TIGER2024/TRACT/"
TRACT_ROOT_DIR = RAW_DIR / "tl_2024_state_tracts"
TRACT_ROOT_DIR.mkdir(parents=True, exist_ok=True)

# derive needed state FIPS directly from tract GEOID
needed_state_fips = sorted(CLUSTER_SCOPE_DF["geoid"].astype(str).str[:2].dro

print("Adjacency mode:", ADJACENCY_MODE)
print("Cluster scope rows:", len(CLUSTER_SCOPE_DF))
print("Needed state FIPS:", needed_state_fips)
print("TRACT_ROOT_DIR:", TRACT_ROOT_DIR)

```

Adjacency mode: shared_edge_or_corner

Cluster scope rows: 213

Needed state FIPS: ['01', '04', '05', '12', '13', '17', '18', '22', '24', '26', '28', '29', '32', '35', '37', '39', '40', '42', '45', '46', '47', '48', '49', '54', '55', '72']

TRACT_ROOT_DIR: data\raw\tl_2024_state_tracts

Cell 21 — load tract geometry

In [171]... *# Cell 21 – download/extract/load standard tract geometry*

```

import geopandas as gpd
from itertools import combinations
from collections import defaultdict, deque

tract_gdf_list = []

for st in needed_state_fips:
    zip_name = f"tl_2024_{st}_tract.zip"
    shp_name = f"tl_2024_{st}_tract.shp"

    zip_path = TRACT_ROOT_DIR / zip_name
    extract_dir = TRACT_ROOT_DIR / f"tl_2024_{st}_tract"
    shp_path = extract_dir / shp_name

    if not shp_path.exists():
        extract_dir.mkdir(parents=True, exist_ok=True)

        if not zip_path.exists():
            url = f"{TRACT_BASE_URL}{zip_name}"
            print(f"Downloading {url} ...")
            r = requests.get(url, timeout=120)
            r.raise_for_status()
            with open(zip_path, "wb") as f:
                f.write(r.content)

```

```

        print(f"Extracting {zip_name} ...")
        with zipfile.ZipFile(zip_path, "r") as zf:
            zf.extractall(extract_dir)

    if not shp_path.exists():
        raise FileNotFoundError(f"Missing shapefile after extraction: {shp_p

    gdf_state = gpd.read_file(shp_path)
    tract_gdf_list.append(gdf_state)

tract_gdf = pd.concat(tract_gdf_list, ignore_index=True)
tract_gdf = gpd.GeoDataFrame(tract_gdf, geometry="geometry", crs=tract_gdf_l

# standard tract GEOID
tract_gdf["geoid"] = normalize_geoid_series(tract_gdf["GEOID"])

tract_gdf = tract_gdf.dropna(subset=["geoid", "geometry"]).copy()
tract_gdf = tract_gdf[tract_gdf.geometry.notna()].copy()
tract_gdf = tract_gdf[~tract_gdf.geometry.is_empty].copy()
tract_gdf = tract_gdf.drop_duplicates(subset=["geoid"]).copy()

print("Combined tract_gdf shape:", tract_gdf.shape)
print("Unique GEOIDs in tract_gdf:", tract_gdf["geoid"].nunique())
print(tract_gdf[["STATEFP", "COUNTYFP", "TRACTCE", "GEOID", "geoid"]].head())

```

Combined tract_gdf shape: (51183, 15)

Unique GEOIDs in tract_gdf: 51183

	STATEFP	COUNTYFP	TRACTCE	GEOID	geoid
0	01	063	060102	01063060102	01063060102
1	01	063	060101	01063060101	01063060101
2	01	069	040802	01069040802	01069040802
3	01	069	040204	01069040204	01069040204
4	01	069	041902	01069041902	01069041902

Cell 22 — merge shortlist with geometry

In [172]... *# Cell 22 – merge shortlist with geometry*

```

adjacency_gdf = tract_gdf.merge(
    CLUSTER_SCOPE_DF,
    on="geoid",
    how="inner",
    validate="1:1"
)

adjacency_gdf = adjacency_gdf[adjacency_gdf.geometry.notna()].copy()
adjacency_gdf = adjacency_gdf[~adjacency_gdf.geometry.is_empty].copy()

print("Adjacency merge shape:", adjacency_gdf.shape)
print("Matched shortlisted tracts with geometry:", adjacency_gdf["geoid"].n
print()

print(adjacency_gdf[
    ["geoid", "display_state", "display_county", "display_name", "igs_total"
].head(10))

```

Adjacency merge shape: (213, 47)
Matched shortlisted tracts with geometry: 213

	geoid	display_state	display_county	display_name	igs_total	total_shortlist_score
0	01101002201	Alabama	Montgomery County	Census Tract 22.01; Montgomery County; Alabama	28.0	83.567419
1	01101003000	Alabama	Montgomery County	Census Tract 30; Montgomery County; Alabama	26.0	84.756545
2	01055000300	Alabama	Etowah County	Census Tract 3; Etowah County; Alabama	25.1	84.642693
3	01025957700	Alabama	Clarke County	Census Tract 9577; Clarke County; Alabama	22.0	84.565702
4	01097000500	Alabama	Mobile County	Census Tract 5; Mobile County; Alabama	24.0	88.467327
5	01097000600	Alabama	Mobile County	Census Tract 6; Mobile County; Alabama	22.0	82.418841
6	01097003902	Alabama	Mobile County	Census Tract 39.02; Mobile County; Alabama	22.0	85.505634
7	01097000701	Alabama	Mobile County	Census Tract 7.01; Mobile County; Alabama	27.0	86.757194
8	01097004100	Alabama	Mobile County	Census Tract 41; Mobile County; Alabama	26.0	84.418318
9	01097001302	Alabama	Mobile County	Census Tract 13.02; Mobile County; Alabama	23.0	87.415197

Cell 23 — build shared-edge/corner adjacency graph

```
In [173... # Cell 23 – build shared-edge/corner adjacency graph

def build_adjacency_edges(gdf: gpd.GeoDataFrame) -> pd.DataFrame:
    """
    Build tract-to-tract adjacency edges within each display_state/display_county.
    Shared edge/corner adjacency uses geometry.touches().
    """
    edge_rows = []

    for (state_name, county_name), grp in gdf.groupby(["display_state", "display_county"]):
        grp = grp.reset_index(drop=True).copy()

        if len(grp) < 2:
            continue

        for i, j in combinations(range(len(grp)), 2):
            geom_i = grp.loc[i, "geometry"]
            geom_j = grp.loc[j, "geometry"]

            if geom_i is None or geom_j is None:
                continue

            if geom_i.touches(geom_j):
                edge_rows.append({
                    "display_state": state_name,
                    "display_county": county_name,
                    "geoid_1": grp.loc[i, "geoid"],
                    "geoid_2": grp.loc[j, "geoid"],
                })
```

```

        "geoid_2": grp.loc[j, "geoid"],
    })

    return pd.DataFrame(edge_rows)

adjacency_edges = build_adjacency_edges(adjacency_gdf)

print("Adjacency edges shape:", adjacency_edges.shape)
print(adjacency_edges.head(20))

```

```

Adjacency edges shape: (91, 4)
   display_state display_county geoid_1 geoid_2
0      Alabama  Mobile County  01097000500  01097000600
1      Alabama  Mobile County  01097000600  01097000701
2      Alabama  Mobile County  01097003902  01097004100
3      Alabama  Mobile County  01097001302  01097001502
4      Alabama  Mobile County  01097001302  01097001501
5      Alabama  Mobile County  01097001502  01097001501
6      Arizona  Apache County  04001942600  04001944301
7      Arizona  Apache County  04001944100  04001944301
8      Arizona  Apache County  04001944301  04001944302
9      Arizona  Apache County  04001944302  04001944902
10     Arizona  Apache County  04001944302  04001944901
11     Arizona  Apache County  04001944902  04001944901
12     Arizona  Apache County  04001945002  04001944901
13     Arizona  Navajo County  04017940012  04017942300
14     Florida  Orange County  12095010400  12095010500
15     Georgia  Bibb County   13021011500  13021012700
16     Georgia  Bibb County   13021012700  13021012800
17     Georgia  Bibb County   13021012800  13021012900
18     Illinois  Cook County   17031681200  17031681300
19     Illinois  Cook County   17031491000  17031491300

```

Cell 24 — connected tract clusters

```

In [174... # Cell 24 — connected tract clusters

def connected_components_from_edges(nodes, edges_df):
    """
    Return a dict: geoid -> cluster_id
    """
    neighbors = defaultdict(set)
    for n in nodes:
        neighbors[n] = set()

    for _, row in edges_df.iterrows():
        a = row["geoid_1"]
        b = row["geoid_2"]
        neighbors[a].add(b)
        neighbors[b].add(a)

    visited = set()
    node_to_cluster = {}
    cluster_id = 0

```



```

for n in nodes:
    if n in visited:
        continue

    cluster_id += 1
    q = deque([n])
    visited.add(n)

    while q:
        cur = q.popleft()
        node_to_cluster[cur] = cluster_id

        for nxt in neighbors[cur]:
            if nxt not in visited:
                visited.add(nxt)
                q.append(nxt)

    return node_to_cluster

all_candidate_nodes = adjacency_gdf["geoid"].astype(str).unique().tolist()
node_to_cluster = connected_components_from_edges(all_candidate_nodes, adjacency_gdf["edges"])

adjacency_gdf["cluster_id_num"] = adjacency_gdf["geoid"].astype(str).map(node_to_cluster)
adjacency_gdf["cluster_id"] = (
    adjacency_gdf["display_state"].astype(str) + " | " +
    adjacency_gdf["display_county"].astype(str) + " | cluster_" +
    adjacency_gdf["cluster_id_num"].astype(str)
)

print("Total cluster members:", len(adjacency_gdf))
print("Unique clusters:", adjacency_gdf["cluster_id"].nunique())
print()

print(adjacency_gdf[
    ["cluster_id", "geoid", "display_state", "display_county", "display_name"]
].head(20))

```

Total cluster members: 213

Unique clusters: 139

	cluster_id	geoid	display_state	d
isplay_county		display_name	total_shortli	
st_score				
0	Alabama Montgomery County cluster_1	01101002201	Alabama	Mont
3.567419	gomery County Census Tract 22.01; Montgomery County; Alabama			8
1	Alabama Montgomery County cluster_2	01101003000	Alabama	Mont
4.756545	gomery County Census Tract 30; Montgomery County; Alabama			8
2	Alabama Etowah County cluster_3	01055000300	Alabama	
4.642693	Etowah County Census Tract 3; Etowah County; Alabama			8
3	Alabama Clarke County cluster_4	01025957700	Alabama	
4.565702	Clarke County Census Tract 9577; Clarke County; Alabama			8
4	Alabama Mobile County cluster_5	01097000500	Alabama	
8.467327	Mobile County Census Tract 5; Mobile County; Alabama			8
5	Alabama Mobile County cluster_5	01097000600	Alabama	
2.418841	Mobile County Census Tract 6; Mobile County; Alabama			8
6	Alabama Mobile County cluster_6	01097003902	Alabama	
5.505634	Mobile County Census Tract 39.02; Mobile County; Alabama			8
7	Alabama Mobile County cluster_5	01097000701	Alabama	
6.757194	Mobile County Census Tract 7.01; Mobile County; Alabama			8
8	Alabama Mobile County cluster_6	01097004100	Alabama	
4.418318	Mobile County Census Tract 41; Mobile County; Alabama			8
9	Alabama Mobile County cluster_7	01097001302	Alabama	
7.415197	Mobile County Census Tract 13.02; Mobile County; Alabama			8
10	Alabama Mobile County cluster_7	01097001502	Alabama	
7.668349	Mobile County Census Tract 15.02; Mobile County; Alabama			8
11	Alabama Russell County cluster_8	01113030300	Alabama	R
5.159427	ussell County Census Tract 303; Russell County; Alabama			8
12	Alabama Jefferson County cluster_9	01073003200	Alabama	Jef
3.599150	fereson County Census Tract 32; Jefferson County; Alabama			8
13	Alabama Greene County cluster_10	01063060000	Alabama	
6.794768	Greene County Census Tract 600; Greene County; Alabama			8
14	Alabama Morgan County cluster_11	01103000600	Alabama	
4.309659	Morgan County Census Tract 6; Morgan County; Alabama			8
15	Alabama Mobile County cluster_12	01097004800	Alabama	
7.440875	Mobile County Census Tract 48; Mobile County; Alabama			8
16	Alabama Mobile County cluster_7	01097001501	Alabama	
	Mobile County Census Tract 15.01; Mobile County; Alabama			8

```

7.657374
17      Arizona | Apache County | cluster_13  04001942600      Arizona
Apache County      Census Tract 9426; Apache County; Arizona      8
0.689922
18      Arizona | Apache County | cluster_13  04001944100      Arizona
Apache County      Census Tract 9441; Apache County; Arizona      8
4.260294
19      Arizona | Apache County | cluster_13  04001944301      Arizona
Apache County      Census Tract 9443.01; Apache County; Arizona      8
8.431911

```

Cell 25 — cluster summaries + serious cluster filter

```

In [175... # Cell 25 - cluster summaries + serious cluster filter

adjacent_cluster_summary = (
    adjacency_gdf
    .groupby(["cluster_id", "display_state", "display_county"], dropna=False)
    .agg(
        n_cluster_tracts=("geoid", "nunique"),
        cluster_total_pop=("pop_total", "sum"),
        mean_shortlist_score=("total_shortlist_score", "mean"),
        max_shortlist_score=("total_shortlist_score", "max"),
        mean_igs_total=("igs_total", "mean"),
        min_igs_total=("igs_total", "min"),
        mean_poverty_rate=("poverty_rate", "mean"),
        mean_insured_share=("insured_share", "mean"),
        mean_unemp_rate=("unemp_rate", "mean"),
        tiny_tract_share=("tiny_tract_flag", "mean"),
        member_geoids=("geoid", lambda s: ", ".join(s.astype(str).tolist())),
        member_names=("display_name", lambda s: " | ".join(s.astype(str).tolist()))
    )
    .reset_index()
    .sort_values(
        ["n_cluster_tracts", "mean_shortlist_score", "cluster_total_pop"],
        ascending=[False, False, False]
    )
    .reset_index(drop=True)
)

serious_adjacent_clusters = (
    adjacent_cluster_summary[
        (adjacent_cluster_summary["n_cluster_tracts"] >= MIN_CLUSTER_TRACTS)
        & (adjacent_cluster_summary["cluster_total_pop"] >= MIN_CLUSTER_POP)
    ]
    .copy()
    .reset_index(drop=True)
)

print("All adjacent clusters:", adjacent_cluster_summary.shape)
print("Serious adjacent clusters:", serious_adjacent_clusters.shape)
print()

print("Top adjacent clusters:")
print(adjacent_cluster_summary.head(25))

```

```
print()

print("Top serious adjacent clusters:")
print(serious_adjacent_clusters.head(25))
```

All adjacent clusters: (139, 15)
 Serious adjacent clusters: (14, 15)

Top adjacent clusters:

	cluster_id	display_state	displ
ay_county n_cluster_tracts cluster_total_pop mean_shortlist_score max_sh	ortlist_score mean_igs_total min_igs_total \		
0 Puerto Rico San Juan Municipio cluster_126	10 20465.0	Puerto Rico	San Juan
Municipio		85.477542	
91.089515	25.910000 22.0		
1 Indiana Lake County cluster_42	7 8138.0	Indiana	La
ke County		84.870445	
88.707871	24.142857 21.0		
2 Arizona Apache County cluster_13	7 24205.0	Arizona	Apac
he County		83.667499	
88.431911	21.000000 0.0		
3 Illinois Cook County cluster_35	6 6651.0	Illinois	Co
ok County		83.756877	
85.943116	24.833333 23.0		
4 Tennessee Shelby County cluster_103	4 6608.0	Tennessee	Shel
by County		87.233203	
91.759100	24.750000 22.0		
5 Puerto Rico San Juan Municipio cluster_127	4 7497.0	Puerto Rico	San Juan
Municipio		85.516860	
87.593411	27.750000 21.0		
6 Georgia Bibb County cluster_24	4 7121.0	Georgia	Bi
bb County		84.610172	
85.227160	26.750000 25.0		
7 Puerto Rico Guánica Municipio cluster_115	4 5736.0	Puerto Rico	Guánica
Municipio		84.261917	
85.605203	24.500000 22.0		
8 Puerto Rico Arecibo Municipio cluster_119	4 9182.0	Puerto Rico	Arecibo
Municipio		83.404809	
85.745110	24.250000 19.0		
9 Ohio Cuyahoga County cluster_88	4 6101.0	Ohio	Cuyaho
ga County		82.975260	
87.098270	24.000000 18.0		
10 Alabama Mobile County cluster_7	3 3849.0	Alabama	Mobi
le County		87.580307	
87.668349	20.333333 18.0		
11 Alabama Mobile County cluster_5	3 4376.0	Alabama	Mobi
le County		85.881121	
88.467327	24.333333 22.0		
12 Illinois Cook County cluster_31	3 7829.0	Illinois	Co
ok County		85.847952	
88.465442	21.666667 19.0		
13 Tennessee Shelby County cluster_99	3 4959.0	Tennessee	Shel
by County		85.279346	
89.833159	24.666667 21.0		
14 Missouri St. Louis city cluster_78	3 3709.0	Missouri	St. L
ouis city		84.579295	
88.718050	23.400000 16.0		
15 Illinois Cook County cluster_34	3 7115.0	Illinois	Co
ok County		84.052074	
88.031018	24.000000 20.0		
16 Ohio Cuyahoga County cluster_87		Ohio	Cuyaho

ga County	3	3710.0	83.389527	
85.918297	26.200000	19.6		
17	Michigan Wayne County	cluster_64	Michigan	Way
ne County	3	5006.0	81.750820	
83.868596	29.666667	28.0		
18	Michigan Wayne County	cluster_58	Michigan	Way
ne County	3	6369.0	81.560396	
82.957358	25.666667	24.0		
19	Arizona Navajo County	cluster_15	Arizona	Nava
jo County	2	4210.0	89.704562	
90.182591	24.000000	24.0		
20	Florida Orange County	cluster_19	Florida	Oran
ge County	2	2080.0	88.034055	
88.857286	25.000000	24.0		
21	West Virginia Ohio County	cluster_112	West Virginia	Oh
io County	2	1592.0	87.012158	
88.797177	24.800000	23.0		
22	Michigan Wayne County	cluster_60	Michigan	Way
ne County	2	2816.0	86.103003	
86.572949	23.500000	22.0		
23	New Mexico San Juan County	cluster_81	New Mexico	San Ju
an County	2	5707.0	85.924402	
87.008022	13.000000	0.0		
24	Puerto Rico Arecibo Municipio	cluster_116	Puerto Rico	Arecibo
Municipio	2	3898.0	85.603129	
87.806225	22.000000	20.0		

	mean_poverty_rate	mean_insured_share	mean_unemp_rate	tiny_tract_share
member_geoids \				
0	0.609223	0.806237	0.188276	0.300000
72127006101, 72127003600, 72127003300, 72127003200, 72127004400, 72127004600, 72127006200, 72127003701, 72127003702,...				
1	0.475419	0.905539	0.182469	0.714286
18089010600, 18089011300, 18089011400, 18089012100, 18089010205, 18089011900, 18089012800				
2	0.392087	0.798777	0.127350	0.142857
04001942600, 04001944100, 04001944301, 04001944302, 04001944902, 04001945002, 04001944901				
3	0.366687	0.804678	0.198619	1.000000
17031670600, 17031670900, 17031670500, 17031671200, 17031671100, 17031834900				
4	0.606949	0.840207	0.315592	0.250000
47157011500, 47157005000, 47157005900, 47157006000				
5	0.663873	0.829843	0.287252	0.500000
72127002800, 72127001400, 72127002900, 72127001302				
6	0.510439	0.861785	0.199660	0.250000
13021011500, 13021012700, 13021012800, 13021012900				
7	0.624630	0.944841	0.254816	0.500000
72055961300, 72055961200, 72055961400, 72055961500				
8	0.579196	0.936063	0.157336	0.250000
72013300500, 72013301600, 72013300200, 72013300400				
9	0.473463	0.897737	0.138813	0.500000
39035120702, 39035119900, 39035120802, 39035121200				
10	0.493845	0.874179	0.191766	0.666667
01097001302, 01097001502, 01097001501				
11	0.344757	0.855686	0.163062	0.666667
01097000500, 01097000600, 01097000701				

12	0.379279	0.858133	0.170257	0.000000
17031681200,	17031681300,	17031710200		
13	0.382263	0.814158	0.149920	0.333333
47157000600,	47157000900,	47157011200		
14	0.331972	0.803498	0.290802	0.666667
29510107200,	29510107400,	29510107500		
15	0.346525	0.851723	0.216731	0.000000
17031491000,	17031491300,	17031530100		
16	0.432516	0.865373	0.135578	0.666667
39035151600,	39035151500,	39035151800		
17	0.399809	0.920866	0.203231	0.333333
26163514100,	26163514300,	26163514500		
18	0.449890	0.790714	0.086639	0.000000
26163526500,	26163525800,	26163526300		
19	0.401241	0.801171	0.210334	0.000000
04017940012,	04017942300			
20	0.520890	0.712983	0.160491	1.000000
12095010400,	12095010500			
21	0.416770	0.838375	0.183604	1.000000
54069000700,	54069000400			
22	0.586050	0.889846	0.064752	1.000000
26163530800,	26163530900			
23	0.425332	0.798328	0.120984	0.000000
35045943100,	35045943000			
24	0.656372	0.948402	0.222265	0.000000
72013302201,	72013302300			

member_names

0 Census Tract 61.01; San Juan Municipio; Puerto Rico | Census Tract 36; San Juan Municipio; Puerto Rico | Census Trac...

1 Census Tract 106; Lake County; Indiana | Census Tract 113; Lake County; Indiana | Census Tract 114; Lake County; Ind...

2 Census Tract 9426; Apache County; Arizona | Census Tract 9441; Apache County; Arizona | Census Tract 9443.01; Apache...

3 Census Tract 6706; Cook County; Illinois | Census Tract 6709; Cook County; Illinois | Census Tract 6705; Cook County...

4 Census Tract 115; Shelby County; Tennessee | Census Tract 50; Shelby County; Tennessee | Census Tract 59; Shelby Cou...

5 Census Tract 28; San Juan Municipio; Puerto Rico | Census Tract 14; San Juan Municipio; Puerto Rico | Census Tract 2...

6 Census Tract 115; Bibb County; Georgia | Census Tract 127; Bibb County; Georgia | Census Tract 128; Bibb County; Geo...

7 Census Tract 9613; Guánica Municipio; Puerto Rico | Census Tract 9612; Guánica Municipio; Puerto Rico | Census Tract...

8 Census Tract 3005; Arecibo Municipio; Puerto Rico | Census Tract 3016; Arecibo Municipio; Puerto Rico | Census Tract...

9 Census Tract 1207.02; Cuyahoga County; Ohio | Census Tract 1199; Cuyahoga County; Ohio | Census Tract 1208.02; Cuyah...

10 Census Tract 13.02; Mobile County; Alabama | Census Tract 15.02; Mobile County; Alabama | Census Tract 15.01; Mobile...

11 Census Tract 5; Mobile County; Alabama | Census Tract 6; Mobile County; Alabama | Census Tract 7.01; Mobile County; ...

12 Census Tract 6812; Cook County; Illinois | Census Tract 6813; Cook County; Illinois | Census Tract 7102; Cook County...

13 Census Tract 6; Shelby County; Tennessee | Census Tract 9; Shelby Count

y; Tennessee | Census Tract 112; Shelby Count...

14 Census Tract 1072; St. Louis city; Missouri | Census Tract 1074; St. Lou
is city; Missouri | Census Tract 1075; St. L...

15 Census Tract 4910; Cook County; Illinois | Census Tract 4913; Cook Count
y; Illinois | Census Tract 5301; Cook County...

16 Census Tract 1516; Cuyahoga County; Ohio | Census Tract 1515; Cuyahoga C
ounty; Ohio | Census Tract 1518; Cuyahoga Co...

17 Census Tract 5141; Wayne County; Michigan | Census Tract 5143; Wayne Cou
nty; Michigan | Census Tract 5145; Wayne Cou...

18 Census Tract 5265; Wayne County; Michigan | Census Tract 5258; Wayne Cou
nty; Michigan | Census Tract 5263; Wayne Cou...

19 Census Tract 9400.12; Navajo County; Ariz
ona | Census Tract 9423; Navajo County; Arizona

20 Census Tract 104; Orange County; Flo
rida | Census Tract 105; Orange County; Florida

21 Census Tract 7; Ohio County; West Virgin
ia | Census Tract 4; Ohio County; West Virginia

22 Census Tract 5308; Wayne County; Michi
gan | Census Tract 5309; Wayne County; Michigan

23 Census Tract 9431; San Juan County; New Mexico |
Census Tract 9430; San Juan County; New Mexico

24 Census Tract 3022.01; Arecibo Municipio; Puerto Rico | Ce
nsus Tract 3023; Arecibo Municipio; Puerto Rico

Top serious adjacent clusters:

	cluster_id	display_state	display
y_county	n_cluster_tracts	cluster_total_pop	mean_shortlist_score
rtlist_score	mean_igs_total	min_igs_total	\
0 Puerto Rico San Juan Municipio	cluster_126	Puerto Rico	San Juan M
unipio	10	20465.0	85.477542
91.089515	25.910000	22.0	
1 Indiana Lake County	cluster_42	Indiana	Lak
e County	7	8138.0	84.870445
88.707871	24.142857	21.0	
2 Arizona Apache County	cluster_13	Arizona	Apach
e County	7	24205.0	83.667499
88.431911	21.000000	0.0	
3 Illinois Cook County	cluster_35	Illinois	Coo
k County	6	6651.0	83.756877
85.943116	24.833333	23.0	
4 Tennessee Shelby County	cluster_103	Tennessee	Shelb
y County	4	6608.0	87.233203
91.759100	24.750000	22.0	
5 Puerto Rico San Juan Municipio	cluster_127	Puerto Rico	San Juan M
unipio	4	7497.0	85.516860
87.593411	27.750000	21.0	
6 Georgia Bibb County	cluster_24	Georgia	Bib
b County	4	7121.0	84.610172
85.227160	26.750000	25.0	
7 Puerto Rico Guánica Municipio	cluster_115	Puerto Rico	Guánica M
unipio	4	5736.0	84.261917
85.605203	24.500000	22.0	
8 Puerto Rico Arecibo Municipio	cluster_119	Puerto Rico	Arecibo M
unipio	4	9182.0	83.404809
85.745110	24.250000	19.0	
9 Ohio Cuyahoga County	cluster_88	Ohio	Cuyahog

a County	4	6101.0	82.975260	
87.098270	24.000000	18.0		
10	Illinois Cook County cluster_31		Illinois	Coo
k County	3	7829.0	85.847952	
88.465442	21.666667	19.0		
11	Illinois Cook County cluster_34		Illinois	Coo
k County	3	7115.0	84.052074	
88.031018	24.000000	20.0		
12	Michigan Wayne County cluster_64		Michigan	Wayn
e County	3	5006.0	81.750820	
83.868596	29.666667	28.0		
13	Michigan Wayne County cluster_58		Michigan	Wayn
e County	3	6369.0	81.560396	
82.957358	25.666667	24.0		

	mean_poverty_rate	mean_insured_share	mean_unemp_rate	tiny_tract_share
member_geoids \				
0	0.609223	0.806237	0.188276	0.300000
72127006101, 72127003600, 72127003300, 72127003200, 72127004400, 72127004600, 72127006200, 72127003701, 72127003702,...				
1	0.475419	0.905539	0.182469	0.714286
18089010600, 18089011300, 18089011400, 18089012100, 18089010205, 18089011900, 18089012800				
2	0.392087	0.798777	0.127350	0.142857
04001942600, 04001944100, 04001944301, 04001944302, 04001944902, 04001945002, 04001944901				
3	0.366687	0.804678	0.198619	1.000000
17031670600, 17031670900, 17031670500, 17031671200, 17031671100, 17031834900				
4	0.606949	0.840207	0.315592	0.250000
47157011500, 47157005000, 47157005900, 47157006000				
5	0.663873	0.829843	0.287252	0.500000
72127002800, 72127001400, 72127002900, 72127001302				
6	0.510439	0.861785	0.199660	0.250000
13021011500, 13021012700, 13021012800, 13021012900				
7	0.624630	0.944841	0.254816	0.500000
72055961300, 72055961200, 72055961400, 72055961500				
8	0.579196	0.936063	0.157336	0.250000
72013300500, 72013301600, 72013300200, 72013300400				
9	0.473463	0.897737	0.138813	0.500000
39035120702, 39035119900, 39035120802, 39035121200				
10	0.379279	0.858133	0.170257	0.000000
17031681200, 17031681300, 17031710200				
11	0.346525	0.851723	0.216731	0.000000
17031491000, 17031491300, 17031530100				
12	0.399809	0.920866	0.203231	0.333333
26163514100, 26163514300, 26163514500				
13	0.449890	0.790714	0.086639	0.000000
26163526500, 26163525800, 26163526300				

member_names

0	Census Tract 61.01; San Juan Municipio; Puerto Rico Census Tract 36; San Juan Municipio; Puerto Rico Census Trac...
1	Census Tract 106; Lake County; Indiana Census Tract 113; Lake County; Indiana Census Tract 114; Lake County; Ind...
2	Census Tract 9426; Apache County; Arizona Census Tract 9441; Apache Co

unty; Arizona | Census Tract 9443.01; Apache...
 3 Census Tract 6706; Cook County; Illinois | Census Tract 6709; Cook Count
 y; Illinois | Census Tract 6705; Cook County...
 4 Census Tract 115; Shelby County; Tennessee | Census Tract 50; Shelby Cou
 nty; Tennessee | Census Tract 59; Shelby Cou...
 5 Census Tract 28; San Juan Municipio; Puerto Rico | Census Tract 14; San
 Juan Municipio; Puerto Rico | Census Tract 2...
 6 Census Tract 115; Bibb County; Georgia | Census Tract 127; Bibb County;
 Georgia | Census Tract 128; Bibb County; Geo...
 7 Census Tract 9613; Guánica Municipio; Puerto Rico | Census Tract 9612; G
 uánica Municipio; Puerto Rico | Census Tract...
 8 Census Tract 3005; Arecibo Municipio; Puerto Rico | Census Tract 3016; A
 recibo Municipio; Puerto Rico | Census Tract...
 9 Census Tract 1207.02; Cuyahoga County; Ohio | Census Tract 1199; Cuyahog
 a County; Ohio | Census Tract 1208.02; Cuyah...
 10 Census Tract 6812; Cook County; Illinois | Census Tract 6813; Cook Count
 y; Illinois | Census Tract 7102; Cook County...
 11 Census Tract 4910; Cook County; Illinois | Census Tract 4913; Cook Count
 y; Illinois | Census Tract 5301; Cook County...
 12 Census Tract 5141; Wayne County; Michigan | Census Tract 5143; Wayne Cou
 nty; Michigan | Census Tract 5145; Wayne Cou...
 13 Census Tract 5265; Wayne County; Michigan | Census Tract 5258; Wayne Cou
 nty; Michigan | Census Tract 5263; Wayne Cou...

Optional Cell 26 — save adjacency outputs

```

In [176... # Cell 26 – save adjacency outputs

if SAVE_INTERMEDIATE:
    safe_save_parquet(adjacency_gdf.drop(columns="geometry"), OUT_DIR / "adj
    safe_save_parquet(adjacent_cluster_summary, OUT_DIR / "adjacent_cluster_
    safe_save_parquet(serious_adjacent_clusters, OUT_DIR / "serious_adjacent

adjacency_gdf.drop(columns="geometry").to_csv(OUT_DIR / "adjacency_cluster_m
adjacent_cluster_summary.to_csv(OUT_DIR / "adjacent_cluster_summary.csv", ir
serious_adjacent_clusters.to_csv(OUT_DIR / "serious_adjacent_clusters.csv",

print("Saved adjacency outputs to:", OUT_DIR)
  
```

Saved adjacency outputs to: data\processed

```

In [177... print(serious_adjacent_clusters.head(25))
  
```

	cluster_id	display_state	display
y_county n_cluster_tracts cluster_total_pop mean_shortlist_score max_sho	rtlist_score mean_igs_total min_igs_total \		
0 Puerto Rico San Juan Municipio cluster_126	Puerto Rico San Juan M		
unipicio 10 20465.0	85.477542		
91.089515 25.910000 22.0			
1 Indiana Lake County cluster_42	Indiana Lak		
e County 7 8138.0	84.870445		
88.707871 24.142857 21.0			
2 Arizona Apache County cluster_13	Arizona Apach		
e County 7 24205.0	83.667499		
88.431911 21.000000 0.0			
3 Illinois Cook County cluster_35	Illinois Co		
k County 6 6651.0	83.756877		
85.943116 24.833333 23.0			
4 Tennessee Shelby County cluster_103	Tennessee Shelb		
y County 4 6608.0	87.233203		
91.759100 24.750000 22.0			
5 Puerto Rico San Juan Municipio cluster_127	Puerto Rico San Juan M		
unipicio 4 7497.0	85.516860		
87.593411 27.750000 21.0			
6 Georgia Bibb County cluster_24	Georgia Bib		
b County 4 7121.0	84.610172		
85.227160 26.750000 25.0			
7 Puerto Rico Guánica Municipio cluster_115	Puerto Rico Guánica M		
unipicio 4 5736.0	84.261917		
85.605203 24.500000 22.0			
8 Puerto Rico Arecibo Municipio cluster_119	Puerto Rico Arecibo M		
unipicio 4 9182.0	83.404809		
85.745110 24.250000 19.0			
9 Ohio Cuyahoga County cluster_88	Ohio Cuyahog		
a County 4 6101.0	82.975260		
87.098270 24.000000 18.0			
10 Illinois Cook County cluster_31	Illinois Co		
k County 3 7829.0	85.847952		
88.465442 21.666667 19.0			
11 Illinois Cook County cluster_34	Illinois Co		
k County 3 7115.0	84.052074		
88.031018 24.000000 20.0			
12 Michigan Wayne County cluster_64	Michigan Wayn		
e County 3 5006.0	81.750820		
83.868596 29.666667 28.0			
13 Michigan Wayne County cluster_58	Michigan Wayn		
e County 3 6369.0	81.560396		
82.957358 25.666667 24.0			

mean_poverty_rate mean_insured_share mean_unemp_rate tiny_tract_share	member_geoids \
0 0.609223 0.806237 0.188276 0.300000	72127006101, 72127003600, 72127003300, 72127003200, 72127004400, 72127004600, 72127006200, 72127003701, 72127003702,...
1 0.475419 0.905539 0.182469 0.714286	18089010600, 18089011300, 18089011400, 18089012100, 18089010205, 18089011900, 18089012800
2 0.392087 0.798777 0.127350 0.142857	04001942600, 04001944100, 04001944301, 04001944302, 04001944902, 0400194500

2, 04001944901				
3	0.366687	0.804678	0.198619	1.000000
17031670600, 17031670900, 17031670500, 17031671200, 17031671100, 17031834900				
4	0.606949	0.840207	0.315592	0.250000
47157011500, 47157005000, 47157005900, 47157006000				
5	0.663873	0.829843	0.287252	0.500000
72127002800, 72127001400, 72127002900, 72127001302				
6	0.510439	0.861785	0.199660	0.250000
13021011500, 13021012700, 13021012800, 13021012900				
7	0.624630	0.944841	0.254816	0.500000
72055961300, 72055961200, 72055961400, 72055961500				
8	0.579196	0.936063	0.157336	0.250000
72013300500, 72013301600, 72013300200, 72013300400				
9	0.473463	0.897737	0.138813	0.500000
39035120702, 39035119900, 39035120802, 39035121200				
10	0.379279	0.858133	0.170257	0.000000
17031681200, 17031681300, 17031710200				
11	0.346525	0.851723	0.216731	0.000000
17031491000, 17031491300, 17031530100				
12	0.399809	0.920866	0.203231	0.333333
26163514100, 26163514300, 26163514500				
13	0.449890	0.790714	0.086639	0.000000
26163526500, 26163525800, 26163526300				

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0	Census Tract 61.01; San Juan Municipio; Puerto Rico Census Tract 36; S an Juan Municipio; Puerto Rico Census Trac...
1	Census Tract 106; Lake County; Indiana Census Tract 113; Lake County; Indiana Census Tract 114; Lake County; Ind...
2	Census Tract 9426; Apache County; Arizona Census Tract 9441; Apache Co unty; Arizona Census Tract 9443.01; Apache...
3	Census Tract 6706; Cook County; Illinois Census Tract 6709; Cook Count y; Illinois Census Tract 6705; Cook County...
4	Census Tract 115; Shelby County; Tennessee Census Tract 50; Shelby Cou nty; Tennessee Census Tract 59; Shelby Cou...
5	Census Tract 28; San Juan Municipio; Puerto Rico Census Tract 14; San Juan Municipio; Puerto Rico Census Tract 2...
6	Census Tract 115; Bibb County; Georgia Census Tract 127; Bibb County; Georgia Census Tract 128; Bibb County; Geo...
7	Census Tract 9613; Guánica Municipio; Puerto Rico Census Tract 9612; G uánica Municipio; Puerto Rico Census Tract...
8	Census Tract 3005; Arecibo Municipio; Puerto Rico Census Tract 3016; A recibo Municipio; Puerto Rico Census Tract...
9	Census Tract 1207.02; Cuyahoga County; Ohio Census Tract 1199; Cuyahog a County; Ohio Census Tract 1208.02; Cuyah...
10	Census Tract 6812; Cook County; Illinois Census Tract 6813; Cook Count y; Illinois Census Tract 7102; Cook County...
11	Census Tract 4910; Cook County; Illinois Census Tract 4913; Cook Count y; Illinois Census Tract 5301; Cook County...
12	Census Tract 5141; Wayne County; Michigan Census Tract 5143; Wayne Cou nty; Michigan Census Tract 5145; Wayne Cou...
13	Census Tract 5265; Wayne County; Michigan Census Tract 5258; Wayne Cou nty; Michigan Census Tract 5263; Wayne Cou...

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